MANAGERIAL ECONOMICS

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Syllabus

0] Fundmental Lecture : Regression

PART 1 :

1] Parametric Microeconomics
   a. Analysis of Demand

2] Production Analysis

3] Cost of Production

4] Revenue Analysis

PART 2 :

5] Market Analysis

6] Objectives of a firm

7] Market Dynamics
   a. Perfect Competition
   b. Imperfect Competition
      - Monopoly
      - Monopolistic competition
      - Oligopoly

Revenue	Income
-Total Money Recieved from selling Goods and it also covers costs. -Sales is important to earn Revenue	-Income is the profit / net income after all costs are subtracted. - No sales required for income
Finance	**Money**
-Management of Money and other financial Resources. - ~~Enables transactions and valuation of goods and se~~ - Enables mobilization and allocation of capital for investment / productive purpose	- Medium of exchange, store of value - Enables transactions and valuation of Goods and services.

- IPO's are for establishing a company, not operating it.
- Credit Gives temporary purchasing power
  Money gives permanent purchasing power

General Knowledge

1907 - A US bank / liquidity panic triggered by a failed stock leading to sharp stock market fall.

1929 - US stock market collapsed after speculative excess, marking the start of the great depression.

1962 - sudden drop due to investor panic and cold war tensions.

1973-74 - Oil crisis crash - oil embargo and inflation caused deep global recession

1987 - Black Monday - Dow plunged 22% in a single day due to computer trading and panic selling.

2000-02 - Dot com Bubble - Overvalued Internet stocks crashed.

2008 - Global financial Crisis - Housing bubble burst and Lehman collapse triggered worldwide recession.

2015 - Chinese Stock Mar. Crash - Overleveraged retail speculation led to massive sell-offs in China.

2020 - Covid 19-Crash - Pandemic fear & lockdown led to fastest markets decline

- New deal Policy - President Franklin D. Roosevelt - to revive the US economy through relief, recovery and reform during great depression by bringing in a lot of money and creating more jobs.

- Monopoly - one seller of a commodity which has no close substitutes.

  TINA - There Is No Alternative

- Monopolistic competition - Many sellers of a commodity having many close substitutes.

OPEC - Organization of Oil Producing Countries

Monopolyst - Monopoly
Monopolystic - like a monopoly

POL - Petrol, Oil, Lubricants

- Stagflation - national income stagnated but infation increased
- Supply - laggered variable - need to Gestate.

LPG - Liberalization, Privatization, Globalization

LPQ - License, Permit, Quota. (Book - 'India Unbound' by Gurucharan Das)

REGRESSION

- Regression converts data clumps into comprehensible information

$$Y = f(X)$$

where, $X \rightarrow$ Regressor (independent / casual variable) - Cause / Stimulant

$Y \rightarrow$ Regressand (dependent / affected Variable) - Effect / Consequence.

$Y = f(X_1, X_2, X_3, \ldots)$ Multi variate

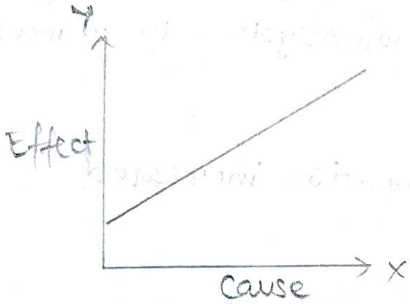
$\frac{dY}{dx} \rightarrow \frac{\text{Effect}}{\text{Cause}}$

* Data should be randomly collected to be scientific.

Positive / Directly Proportional

$X \propto Y$ If, $X \uparrow$, $Y \uparrow$ and
If $X \downarrow$, $Y \downarrow$

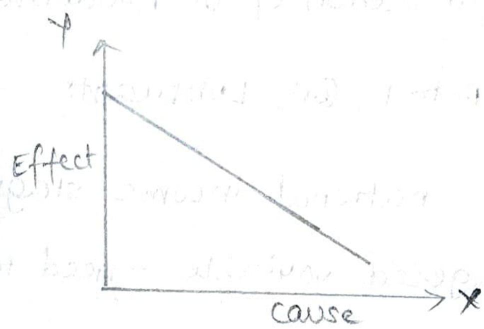
$X\uparrow, Y\uparrow$: $\frac{+ve}{+ve}$ | $X\downarrow\ Y\downarrow$: $\frac{-ve}{-ve}$ } Both the cases give +ve results.



Negative / Indirectly Proportional

$X \propto \frac{1}{Y}$ If, $X \uparrow$, $Y \downarrow$
If $X \downarrow$, $Y \uparrow$

$X\uparrow\ Y\downarrow$: $\frac{+ve}{-ve}$ | $X\downarrow\ Y\uparrow$: $\frac{-ve}{+ve}$ } Both the case give -ve results.



- Slopedness shows Elasticity,
  Trajectory shows direction and,
  Value of Slope shows rate of change.

  $Y = mX + h$ OR $Y = h + mX$

  Linear equation so highest power of eq. is 1.

- Population Regression Function (PFR) → N
  Sample Regression Function (SFR) → n

  * $\boxed{n = 10\%\ N \quad n > 30}$

- Significance of h : Effect takes place in absence of cause.

X	Y	x	y	xy	x^2	$\hat{Y}$	e $(Y-\hat{Y})$	e^2	y^2
1	17	-2	-3	6	4	21	-4	16	9
2	20	-1	0	0	1	20.5	-0.5	0.25	0
3	27	0	7	0	0	20	7	49	49
4	23	1	3	3	1	19.5	3.5	12.25	9
5	13	2	-7	-14	4	19	-6	36	49
15	100	0	0	-5	10				116

$\overline{X}$ (mean of X) $= \frac{15}{5} = 3$

$\overline{Y}$ (mean of Y) $= \frac{100}{5} = 20$

$x = X - \overline{X}$

$y = Y - \overline{Y}$

$$m = \frac{\Sigma xy}{\Sigma x^2} = \frac{-5}{10} = -0.5$$

$h = \overline{Y} - \overline{X}(m)$

$= 20 - 3(-0.5)$

$= 20 + 1.5 = 21.5$

$\hat{Y} = 21.5 - 0.5(X)$

1) $\hat{Y} = 21.5 - 0.5(1) = 21$

2) $\hat{Y} = 21.5 - 0.5(2) = 20.5$

3) $\hat{Y} = 21.5 - 0.5(3) = 20$

4) $\hat{Y} = 21.5 - 0.5(4) = 19.5$

5) $\hat{Y} = 21.5 - 0.5(5) = 19$

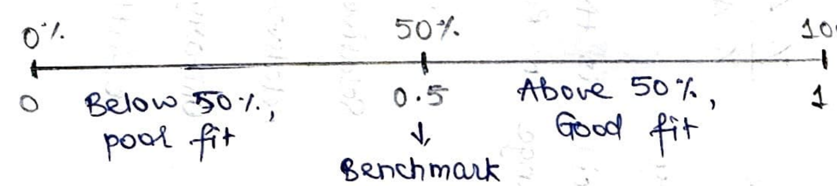
$$R^2 = \frac{(m^2)(\Sigma x^2)}{(\Sigma y^2)}$$

$$= \frac{(-0.5)^2\,(10)}{116}$$

$= 0.02 = 2\%$

R^2 – coefficient of Goodness of fit.

$0 \leq R^2 \leq 1$



$$e_{YX} = \frac{\%\,\Delta Y}{\%\,\Delta X} = \left(\frac{\Delta Y}{\Delta X}\right)\left(\frac{X}{Y}\right) = m\left(\frac{X}{Y}\right) = \tan\theta\left(\frac{X}{Y}\right)$$

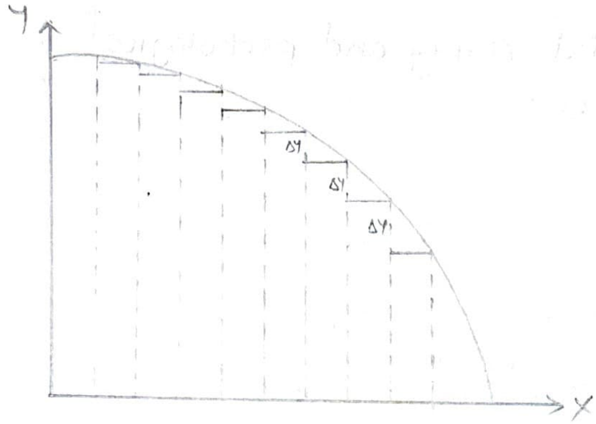
↑
elasticity

Elasticity is the ratio of % change in Regressand (effect) to the % change in Regressor (cause)

Relatively Elastic	$\%\,\Delta Y > \%\,\Delta X$	$e > 1$
Relatively Inelastic	$\%\,\Delta Y < \%\,\Delta X$	$e < 1$
Unitary Elastic	$\%\,\Delta Y = \%\,\Delta X$	$e = 1$
Perfectly Inelastic	$\%\,\Delta Y = 0$ and $\%\,\Delta X = 0$	$e = 0$

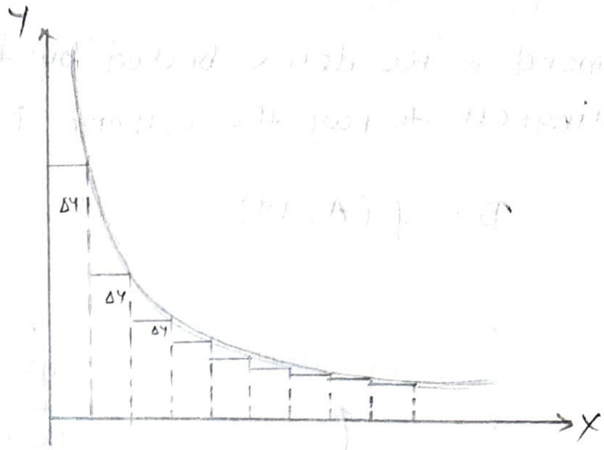
CONCAVITY, CONVEXITY AND INFLEXION

Concave to X axis downwards



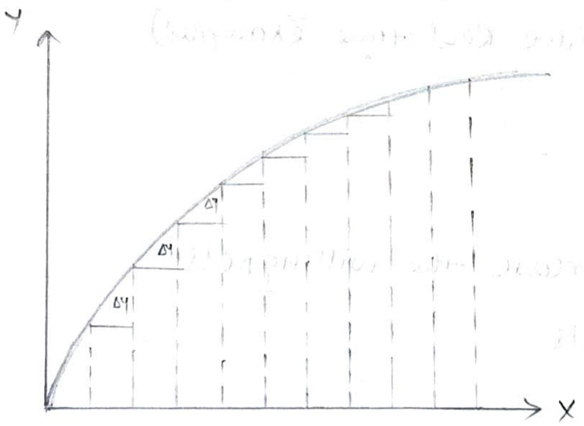
- ΔY goes on increasing
- For more and more of X, we give up or trade off more and more of Y.
- Slower pace - increasing.

Convex to X axis downwards



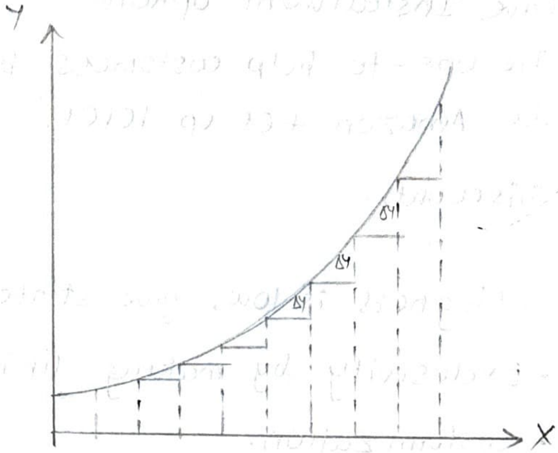
- ΔY goes on decreasing
- For more and more of X, we give up / trade-off ~~more and more~~ less and less of Y.
- Faster pace - decreasing.

Concave to X-axis upwards



- ΔY goes on decreasing
- For more and more of X, we get less and less additions of Y.
- Slower pace

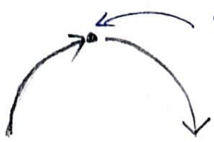
Convex to X axis upwards.



- ΔY goes on increasing
- For more and more of X, we get more and more additions of Y.
- Faster pace.

<u>Inflexion</u> : Point at which 'Rate of Change' (slope) changes, without changing the direction.

inflexion point

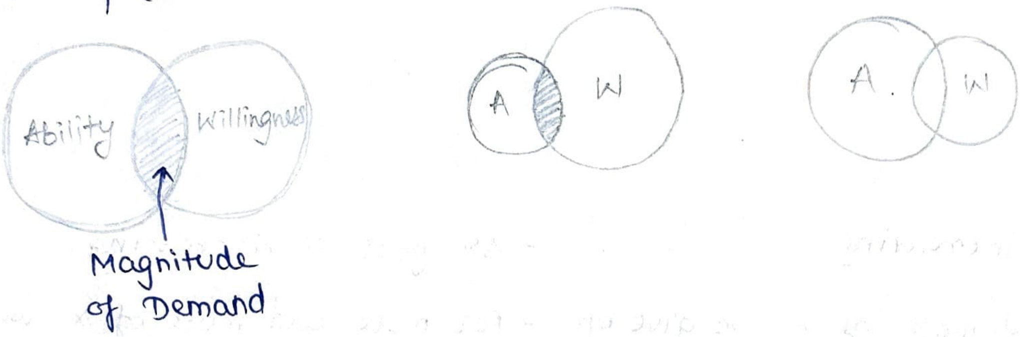


No inflexion.

Conceptualized demand :

- Demand is the desire backed by the financial Ability and pschological willingness to pay the cullent Market price.

$$D = f(A, W)$$



Q] If Ability is lowel than willingness, Give strategies to increase the ability. (Fol particulal client gloup)

→ • Give Installment options
• Tie ups - to help customels buy mole (Give Real time examples) like Amazon ties up ICICI.
• Discounts.

Q] If willingness is low, give strategies to increase the willingness.

→ • Exclusivity by making limited products
• Customization.

EVERY POINT SHOULD GIVE EXAMPLE (Real life)

ELASTICITY OF DEMAND

It is the responsiveness of demand for good X in response to price of X, income of prospective consumer, price of related goods (substitutes, complements), promotional expenditures.

e_{dp} - price elasticity of demand

effect ↗ ↖ cause

$$e_{dp} = \frac{\text{\% change in demand}}{\text{\% change in 4 causes}}$$

Stimuli	Technical Term	Symbol
Price of Concerned Good	Price e_d	e_{dp}
Income of potential consumers	Income e_d	e_{dY}
Price of Related Goods	Cross e_d	e_{dXY}, Good X, Price of good Y
Promotions	Promotional e_d	e_{dA}

$$e_{dp} = \frac{\% \Delta D}{\% \Delta P}$$

(% is very imp. bcoz on $\frac{\Delta D}{\Delta P}$ will ~~get~~ require units)

$$= \frac{\frac{\Delta D}{D_0} \times 100}{\frac{\Delta P}{P_0} \times 100} = \frac{\Delta D}{D_0} \times \cancel{100} \times \frac{P_0}{\Delta P} \times \frac{1}{\cancel{100}}$$

$$= \frac{\Delta D}{D_0} \times \frac{P_0}{\Delta P} = \left(\frac{\Delta D}{\Delta P}\right) \cdot \left(\frac{P_0}{D_0}\right)$$

$$e_{dp} = m\left(\frac{P}{D}\right) = \tan A\left(\frac{P}{D}\right)$$

5 degrees of Elasticity :

1) Perfectly Elastic
2) Relatively Elastic
3) Unitary Elastic
4) Relatively Inelastic
5) Perfectly Inelastic

Demand And Degrees of Elasticity :

1) $\% \Delta P \rightarrow \% \Delta D = 0$ Perfectly Inelastic Demand $e_{dp} = 0$

Eg: Pharma companies - they know customer needs their products and they are dealing with life and death.

2) $\% \Delta P > \% \Delta D$ Relatively Inelastic Demand $e_{dp} < 1$

Eg: Necessities like salt, sugar Change in numer. < Change in Denominator.

3) $\% \Delta P = \% \Delta D$ Unitary Elastic Demand $e_{dp} = 1$

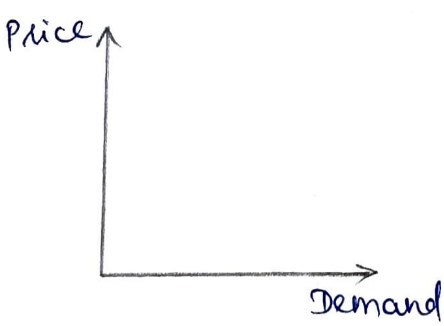
No Real life example - only in theory.

4) $\% \Delta P < \% \Delta D$ Relatively Elastic Demand $e_{dp} > 1$

Eg: Luxuries

5) $\% \Delta P \rightarrow \% \Delta D = \infty$ Perfectly Elastic Demand $e_{dp} = \infty$

Marshallian Model :



Price → cause
Demand → effect

Key features of this Model :

- Incorporate time element (market, short run, long run)
- Role of Elasticity in price adjustments.

① Price Elasticity of Demand:

$$e_{dp} = \frac{\% \Delta D_x}{\% \Delta P_x} = \left(\frac{\Delta D_x}{\Delta P_x}\right)\left(\frac{P_{x_0}}{D_{x_0}}\right) = m\left(\frac{P_{x_0}}{D_{x_0}}\right) = \frac{\text{Lower seg.}}{\text{Upper seg.}}$$

* Tabular data - use % formula ①

Equational data - use slope formula. ③

Q1.

P_x	D_x
10	400
20	330

→ $\% \Delta D_x = \frac{-70}{400} \times 100 = -17.5$

$\% \Delta P_x = \frac{10}{10} \times 100 = 100$

$\therefore e_{dp} = \frac{-17.5}{100} = -0.175 = |0.175|$

$\because e_{dp} = |0.175| < 1$, it shows Relatively Inelastic Demand.

Q2.

P_x	D_x
25	550
10	880

$\% \Delta D_x = \frac{880 - 550}{550} \times 100 = 60$

$\% \Delta P_x = \frac{10 - 25}{25} \times 100 = -60$

$\therefore e_{dp} = \frac{60}{-60} = -1 = |1|$ .... Unitary Elastic Demand.

Q3. $D_x = 120 - 2.7 P_x$; $P_x = 25$ $e_{dp} = ?$

→ $m = -2.7$

$D_x = 120 - 2.7(25)$

$= 52.5$

using $m\left(\frac{P_{x_0}}{D_{x_0}}\right) = -2.7\left(\frac{25}{52.5}\right) = -1.289 = |1.289|$

.... Relatively Elastic

Interpretation :

RIE is good when price rises (demand falls but not much)
RIE is bad when price falls (even though price drops, demand too drops a bit)

RE is bad when price rises (market is emptying)
RE is good when price falls (market is increasing)

(2) Income Elasticity of Demand :

$$e_{dY} = \frac{\% \Delta D_x}{\% \Delta Y} = \left(\frac{\Delta D_x}{\Delta Y}\right)\left(\frac{Y_0}{D_{x_0}}\right) = m\left(\frac{Y}{D_x}\right)$$

Q1.

Income	Demand
10,000	400
25,000	600

$e_{dY} = \frac{50}{150} = 0.33$ RIE

* RIE is bad when income increases.

Q2.

Income	Demand
1,00,000	600
75,000	400

$\% \Delta D_x = \frac{-200}{600} \times 100 = -33.33\%$

$\% \Delta Y = \frac{-25\cancel{000}}{1\cancel{00000}} \times 1\cancel{00} = -25\%$

$\therefore e_{dY} = \frac{\cancel{-}33.33\%}{\cancel{-}25\%} = 1.33$ RE

Q3. $D_x = 7000 + 0.3Y$; $Y = 20,000$; $e_{dY} = ?$

$\rightarrow D_x = 7000 + \frac{3}{1\cancel{0}} \times 2000\cancel{0}$

$= 13000$

using $m\left(\frac{Y}{D_x}\right) = \frac{3}{1\cancel{0}}\left(\frac{20,00\cancel{0}}{13,000}\right) = \frac{6}{13} = 0.46$ RIE

Q] How do you convert unfavourable e_{dp} to favourable e_{dp}?

OR.

How do you convert bad cream batch to good cream batch?

→ • Offer discounts - but not a good strategy to follow when prices are falling because if prices are already falling, there is no need to give discounts. It creates a doubt if the products are faulty.

• Brand Recreation - Eg: Parle to Parle Gold - it is willingness and does not affect the ability.

• Creating exclusivity in products - 'only 5 in whole world' etc. (willingness)

• Offer free services - affects ability.

• Provide EMI's - affects ability.

Disposable Income ⇒ Income you bring home after Tax.

Increase in income is not necessarily increase in salary.

RE is favourable when purchasing power increases and RIE is bad.

Q] How do you make unfavourable e_{dy}, favourable?

Interest rates decrease - RIE

• Offer EMIs

• Offer discounts, offers, bonus referrals

• Card discounts

• free services.

③ Promotional Elasticity of Demand :

$$e_{dA} = \frac{\% \Delta D_x}{\% \Delta A} = \left(\frac{\Delta D_x}{\Delta A}\right)\left(\frac{A_0}{D_{x0}}\right) = m\left(\frac{A}{D_x}\right)$$

Q1.

Prom. Exp.	Demand
50	400
80	900

$\% \Delta D_x = \frac{500}{400} \times 100 = 125\%$

$\% \Delta A = \frac{30}{50} \times 100 = 60\%$

$\therefore e_{dA} = \frac{125\%}{60\%} = 2.08$

Relative Elastic

Q2. $D_x = 10 + 0.7A \quad A = 50 \quad e_{dA} = ?$

$D_x = 10 + \frac{7}{10}(50) \quad m = 0.7$

$= 45$

using, $m\left(\frac{A}{D_x}\right) = \frac{7}{10}\left(\frac{50}{45}\right) = \frac{7}{9} = 0.77$ Relative Inelastic demand

* If any value is less than 0.5 (RIE), campaign is flop.

④ Cross Elasticity of Demand :

$$e_{dxy} = \frac{\% \Delta D_x}{\% \Delta P_y} = \left(\frac{\Delta D_x}{\Delta P_y}\right)\left(\frac{P_{y0}}{D_{x0}}\right) = m\left(\frac{P_y}{D_x}\right)$$

Competitive / Substitute / Rival Goods

$\uparrow P_y \dashrightarrow \downarrow D_y \dashrightarrow \uparrow D_x$

Positive Relationship (look for 1st & last arrow)

P_y	D_x
50	400
80	900

$\% \Delta D_x = \frac{500}{400} \times 100$

$= 125$

$\% \Delta P_y = \frac{30}{50} \times 100 = 60$

$e_{dxy} = \frac{125}{60} = 2.08$

Supportive / Complementary Goods.

$\uparrow P_y \dashrightarrow \downarrow D_y \dashrightarrow \downarrow D_x$

Negative Relationship

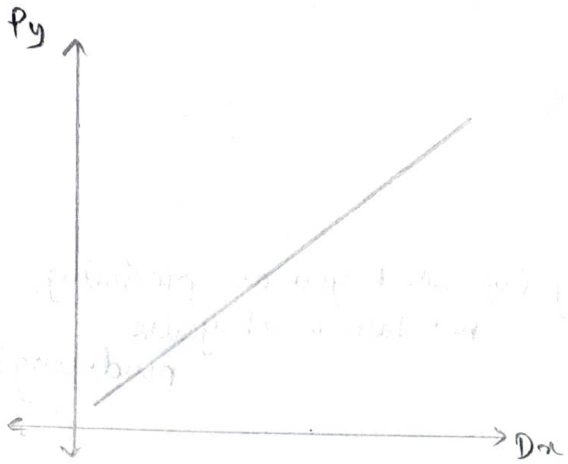
P_y	D_x
50	900
80	400

$\% \Delta D_x = \frac{-500}{900} \times 100$

$= -55.6$

$\% \Delta P_y = \frac{30}{50} \times 100 = 60$

$e_{dxy} = \frac{-55.6}{60} = -0.92$ RIE

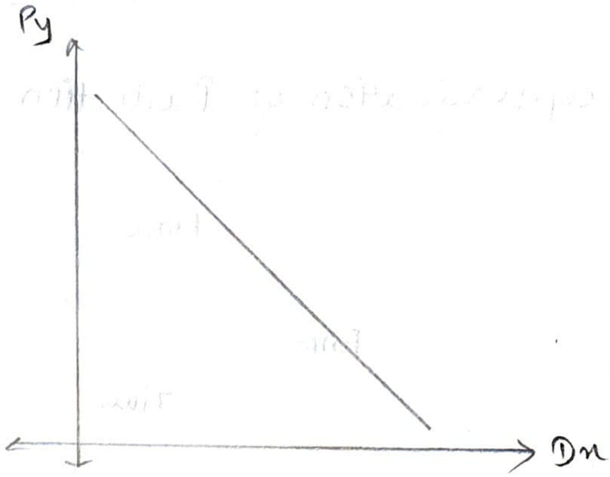


- TINA factor - There is No Alternative

- Beneficial for good x because price of Y increases, demand decreases ∴ Demand of x increases.

- If there is a TINA factor,
  $D_x = 0$
  $e_{dxy} = 0$, market is monopolized. Perfectly inelastic

- When there is no TINA factor, there is competition.



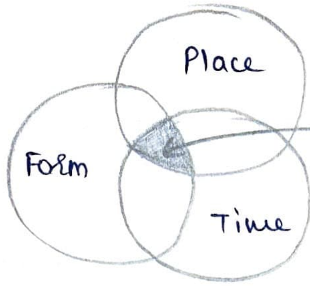
- If RIE and price of relative good is high it is good for the other 'Good' because both are not losing market significantly - effect is insignificant.

- If it is RE then bad because both are losing market significantly

Eg: • Logitech produces supportive goods and ∴ depends on main goods.
• Supportive goods constantly need to be innovated in order to survive when main good is getting a beating.

PRODUCTION

Conceptualization of Production :



Efficiency (of what you are producing, not how much you're producing)

The bigger any circle, the higher it's importance.

Eg: Coke - Form (water) shall be biggest followed by time and place shall be ~~be~~ the smallest.

Production : Transformation of utility in terms of Form, Place, and Time.

Measurement of Production :

1) Total Physical Production (TPP) - refers to overall physical production due to the quantity of inputs used.

$$\boxed{TPP_n = \int_1^n (MPP)(dQ)}$$

Quantity of input changes.

Marginal phy. production

2) Average physical production (APP) - refers to the TPP produced per unit of the inputs used.

$$\boxed{APP_n = \frac{TPP_n}{Q_n}}$$

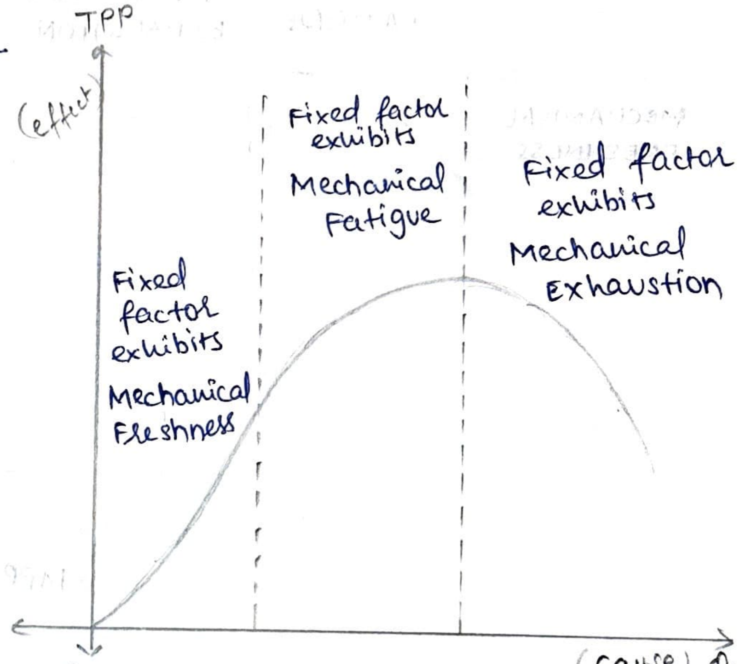
3) Marginal physical production (MPP) - refers to addition caused to TPP by using additional ~~input~~ units of inputs.

$$\boxed{MPP_n = \frac{d(TPP_n)}{d(Q_n)}} \quad \text{OR} \quad \boxed{MPP_n = TPP_n - TPP_{(n-1)}}$$

Geometry of a derivative is Tangents.

Calculation of TPP, APP ~~and~~ given MPP

Q_L	TPP	APP	MPP
1	50	50	50
2	110	55	60
3	180	60	70
4	260	65	80
5	320	64	60
6	360	60	40
7	360	52	0
8	350	44	-10
9	330	37	-20
10	300	30	-30



(Graph is made based on the MPP figures)

Maxima	Minima	Stationality
$f'(x) = 0$	$f'(x) = 0$	$f''(x) = 0$
$f''(x) < 0$	$f''(x) > 0$	

Peak is a point after which there is a fall

- Marginal shapes the total.
- when T is at peak (maxima), M becomes zero.

Law of Variable Proportions / Short term Law of Production / Law of Returns to a factor

Q] Equivalence of All 3 names.

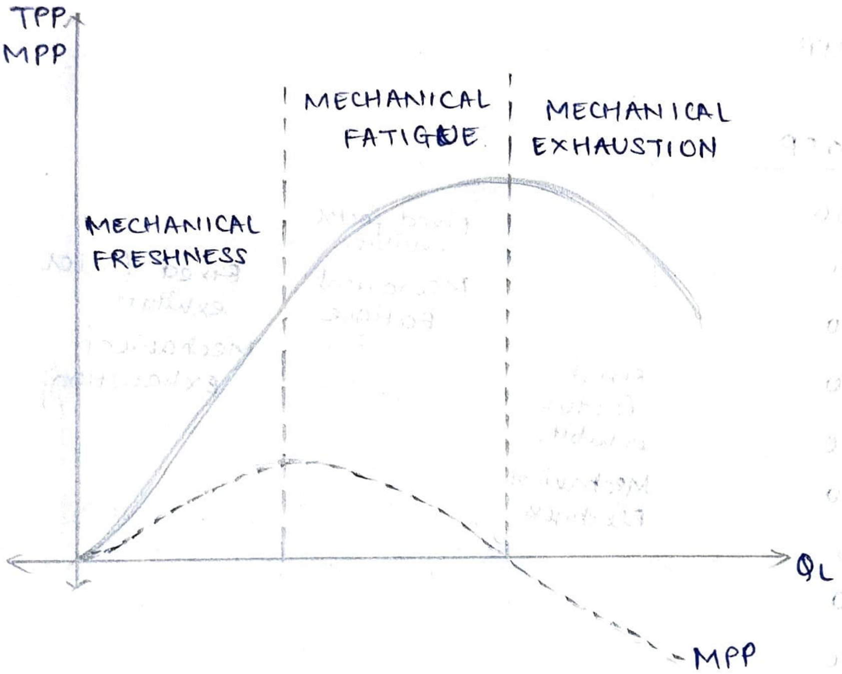
→ (In terms of FFE):

- Short term Law - in which technology does not change
- Law of Returns to a factor - Relates to how the fixed factor is performing.
- Law of Variable proportions - Variable is 'M' (marginal)

Variability is marginal productivity of variable

Proportion is $\frac{\text{1 fixed factor}}{\text{No. of input quantity}} \Rightarrow \frac{\text{Fixed factor}}{\text{Variable factor}}$

Commercial use of Each Stage :



1) Mechanical Freshness -
- Determine warranty and Guarantee period of Machine
- Maintenance business flourishing.

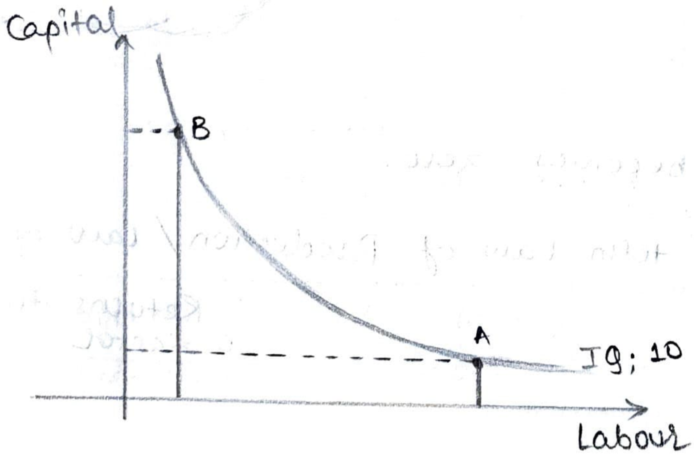
2) Mechanical Fatigue -
- Repair / Replace / Refurbish
- Rate of discount.

3) Mechanical Exhaustion -
- Recycle / Salvage / Scrap.

Long Term Law :

Ⓐ Isoquant Analysis / Equal Quantity Analysis

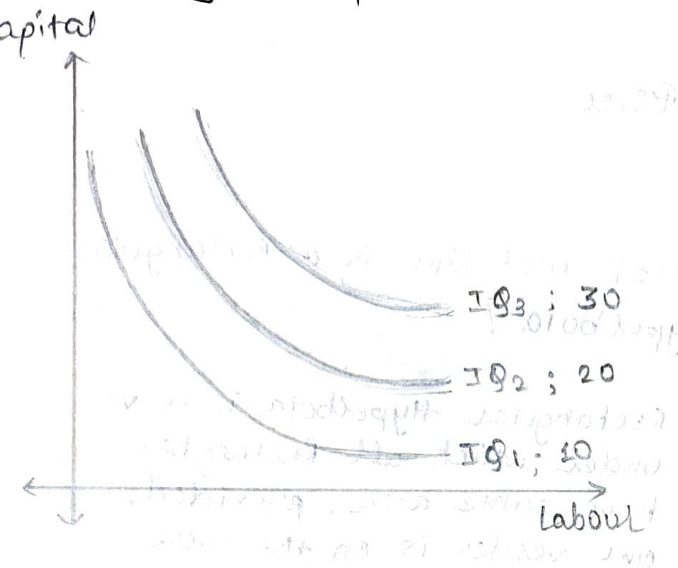
Innovation - New way of popularizing
Technology changes due to inventions and sustains due to innovation.



Locus - function having similar properties.

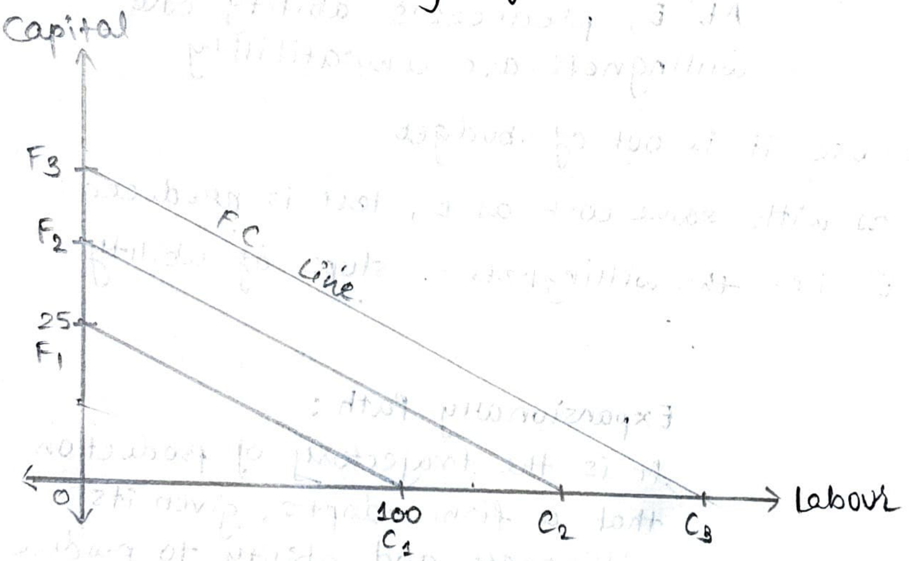
- Isoquant is locus of varying quantities of L and capital (K) showing the same level of output.

* Willingness of Producer - Isoquant Map.



Isoquant Map refers to the willingness of the producer to produce a certain quantity of output.

* Financial Ability of Producer - Factor cost Line.

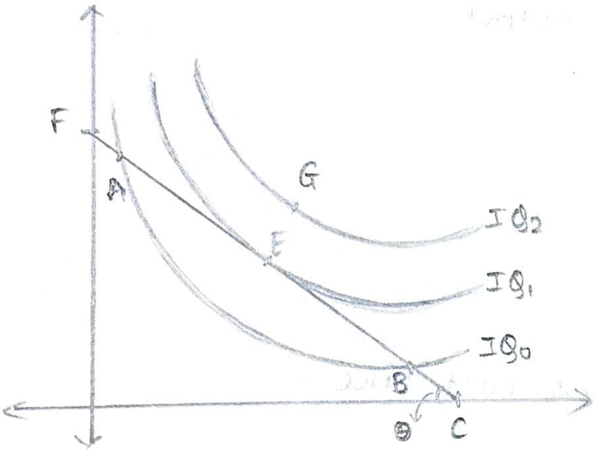


- We assume that the firm's budget is ₹5000 and the cost of labour is ₹50/unit. and the cost of capital is ₹200/unit.
- If the firm spent the entire budget only on labour, the Labour(L) - Capital (K) [L, K] coordinates would be (100L, zero K)
- If entire budget is spent only on capital, the L-K coordinates would be (zero L, 25K)
- If these extreme coordinates when plotted gives us the firm's Factor cost Line.
- Suppose the firm's budget increases to ₹7000 and the price of L and K does not change, the factor cost Line will shift upwards parallelly.

$$\text{Price} = \frac{\text{Expenditure}}{\text{Quantity}}.$$

★ Slope of Factor Cost Line is the Price

Producer's Equilibrium

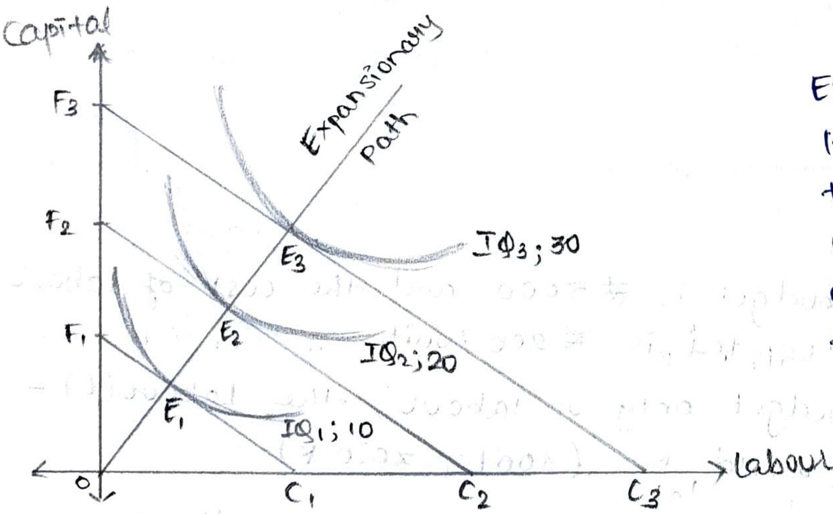


Factor cost Line is a Rectangular Hyperbola ?

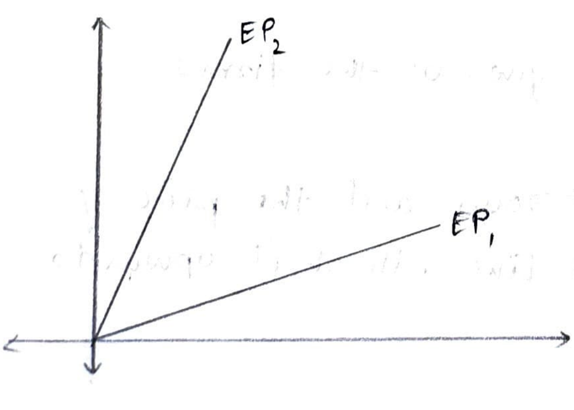
Rectangular Hyperbola is curve under which all Rectangles have same area, provided, one vertex is on the curve

- E is the highest production point At E, producer's ability and willingness are compatibility

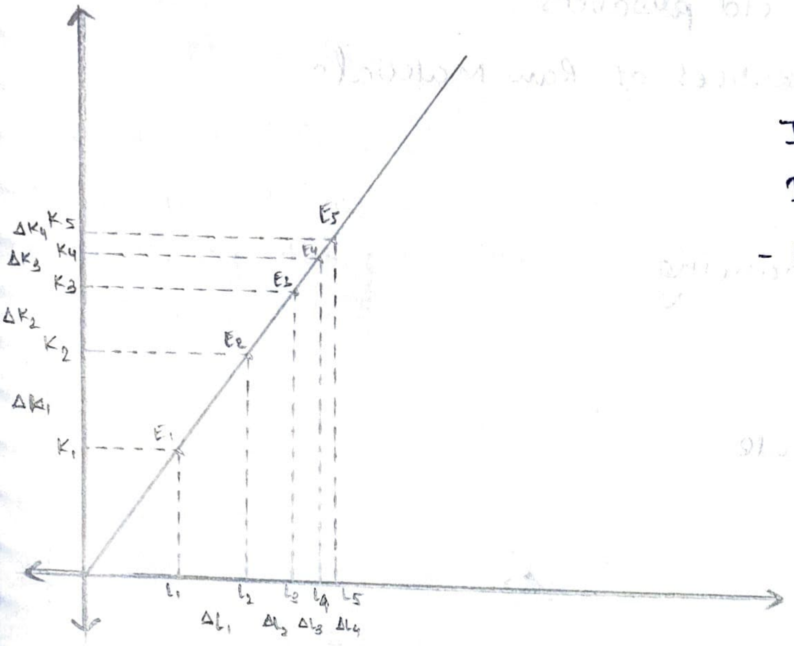
- G is aspirational because it is out of budget.
- A and B are irrational as with same cost as E, less is produced
- θ is the tangent for E i.e. the willingness = slope of ability.



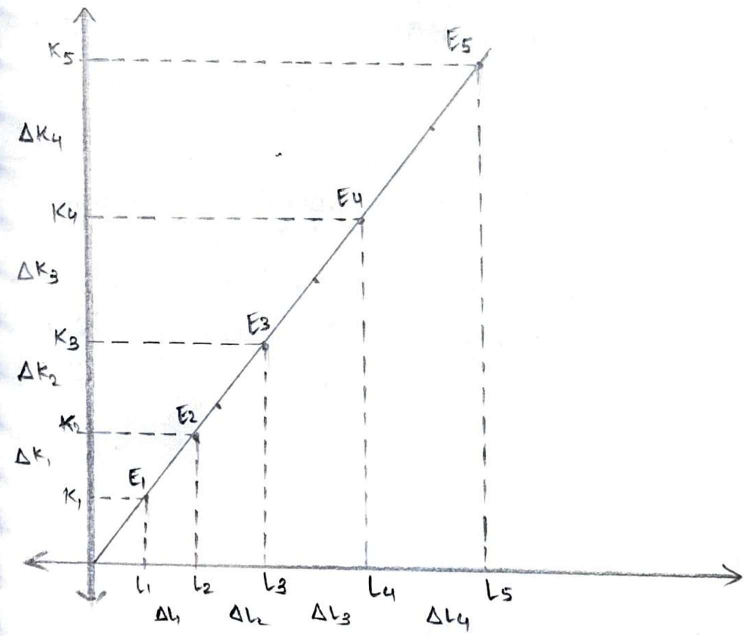
Expansionary Path :
It is the trajectory of production that a firm adopts, given it's willingness and ability to produce as well as the state of technology..



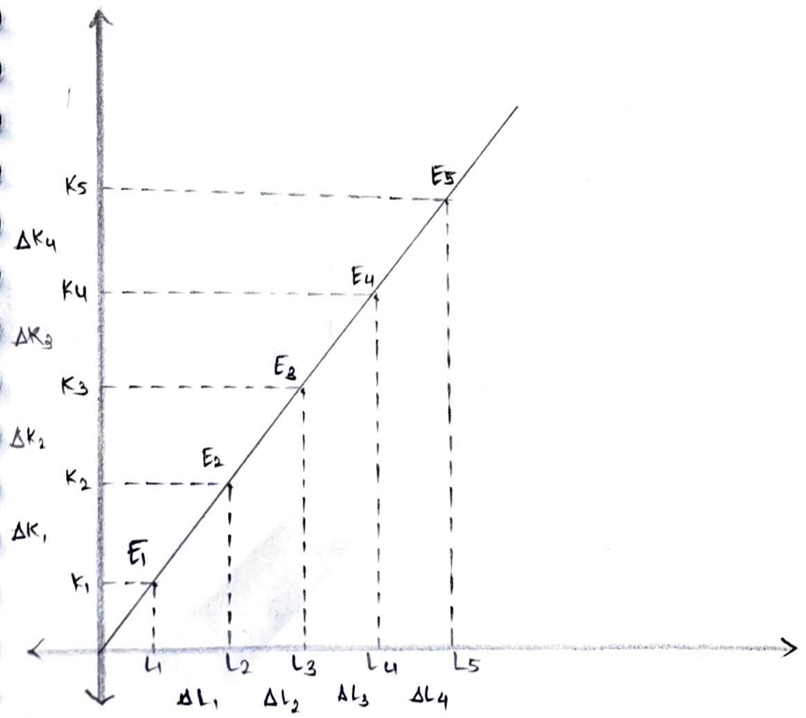
Ⓑ Long Term Law of Productions / Law of Returns to Scale :



Increasing R.S.
Decreasing Gaps
- Expenditure is less to produce same output.



Decreasing R.S.
Increasing Gaps
- Expenditure is more to produce same output.



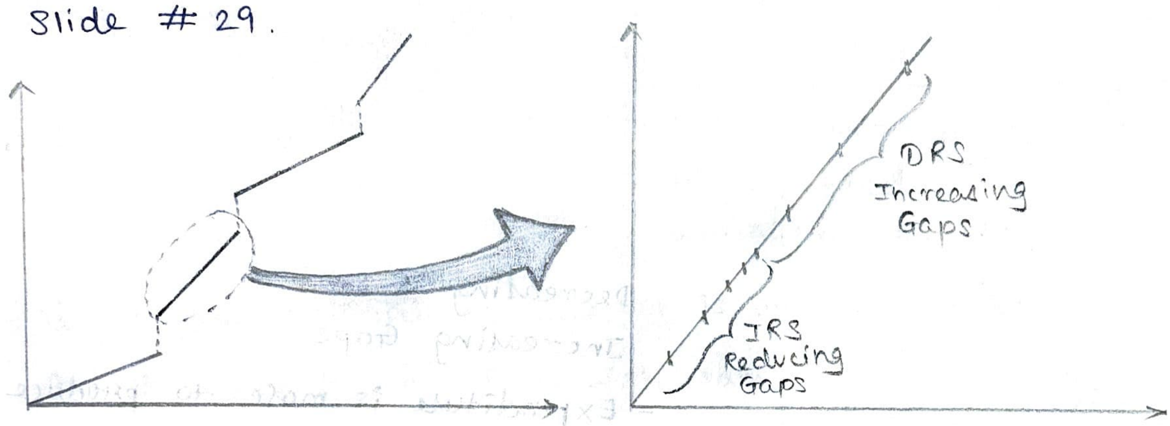
Constant R.S.

Schumpeter's Innovation :-

1) New products, new versions of old products
2) New raw materials and new sources of Raw Materials
3) New Methods of Production
4) New markets
5) New methods of business or financing.

Q] Discuss Laws of Return to Scale.

→ Slide # 29.



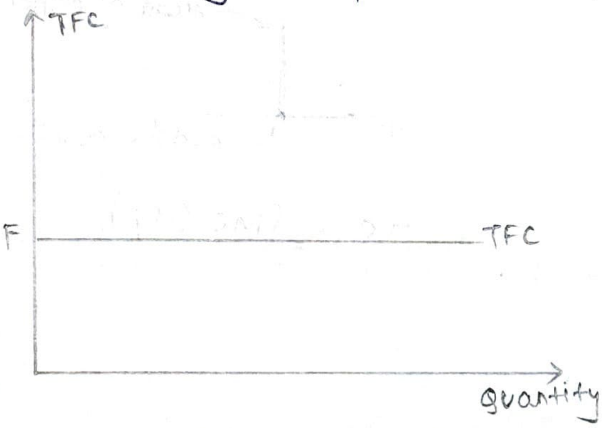
Q] Explain IRS, DRS.

COST OF PRODUCTION

Short Run Cost

Fixed Costs	Variable Costs

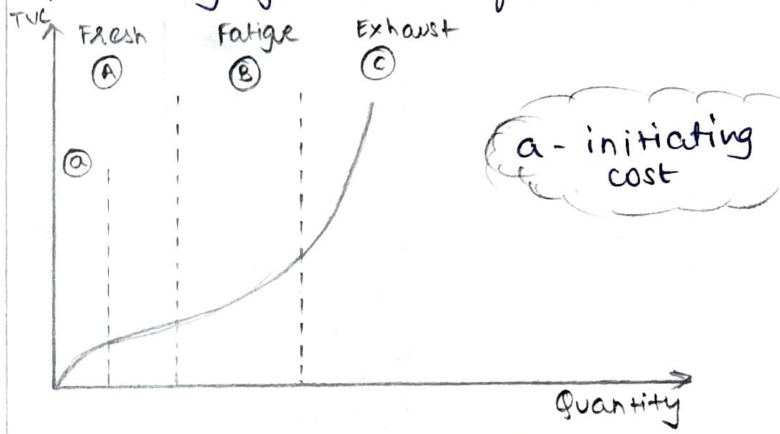
- Total Fixed Cost (TFC)
- Total initial cost incurred in establishing the productive capacity.



$TFC = AFC \times Quantity$

$TFC \neq f(Q)$

- Total Variable Cost (TVC)
- Total operational cost incurred in producing given level of output.

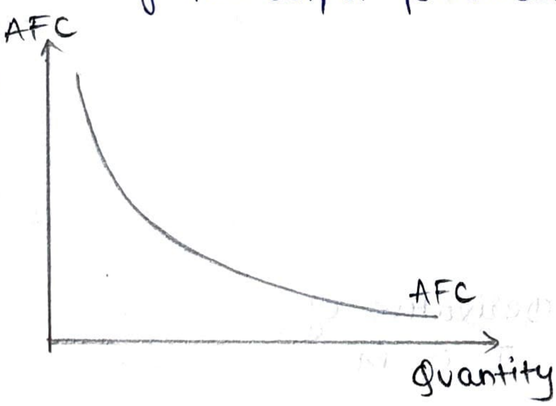


$TVC = AVC \times Quantity$

$TVC = f(Q)$

Average Fixed Cost (AFC)

- It is total fixed cost incurred per unit of the output produced.



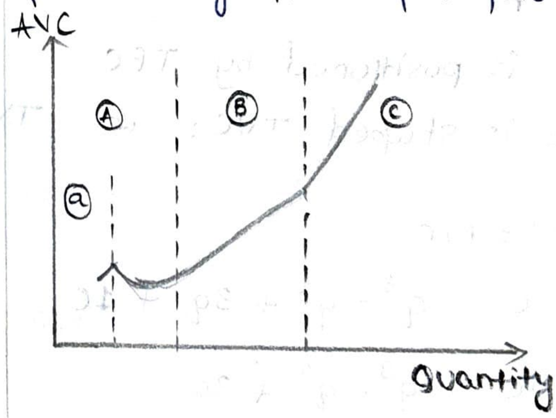
$$AFC = \frac{TFC}{Q}$$

$AFC = f(Q)$

- AFC curve features are :
  - A Rectangular Hyperbola
  - Asymptotic - AFC cannot be zero because the numerator can never be zero. Also, Qty cannot be zero coz of mechanical expiration, production stops.

Average Variable Cost (AVC)

- It is total variable cost incurred per unit of the output produced.

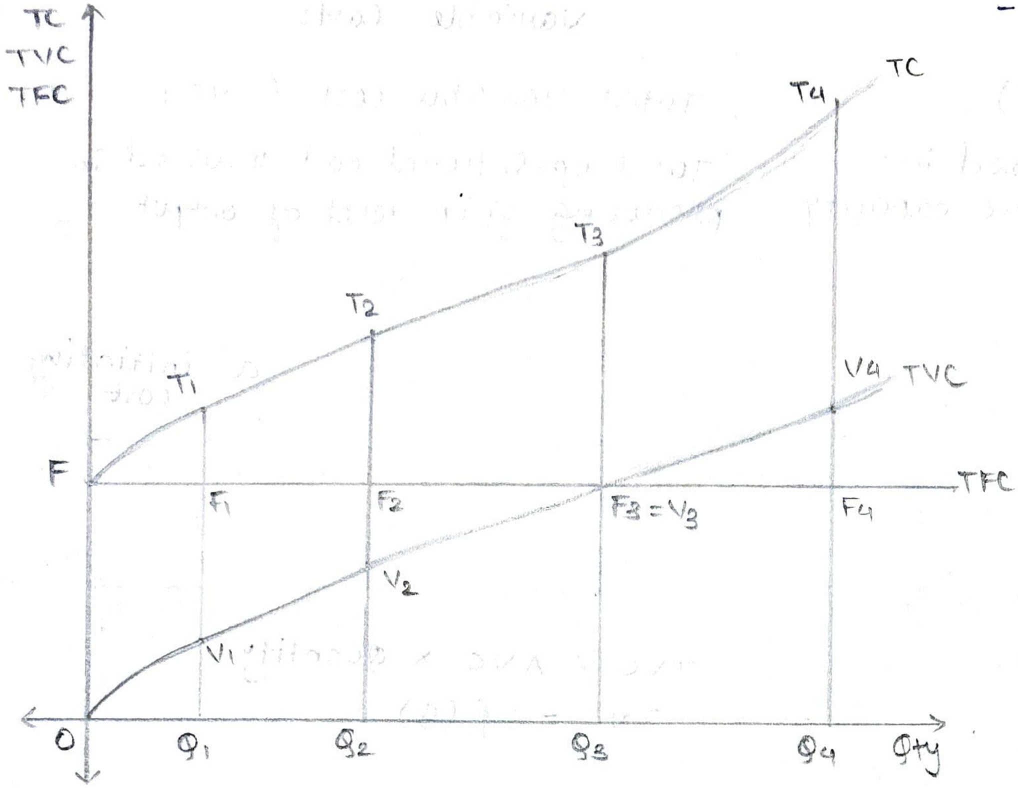


$$AVC = \frac{TVC}{Q}$$

$AVC = f(Q)$

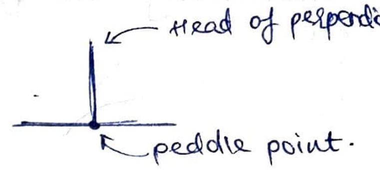
- AVC curve is nearly J shaped.

3 Total Cost



- It is overall cost incurred in producing a given output level.

TC = TFC + TVC



$TC = \int MC \, (dq)$

- TC curve shares the vertical intercept with TFC i.e. TC curve starts at point F.
- TC curve is positioned by TFC
- TC curve is shaped TVC; i.e. TVC' = MC

TVC' = MC

<u>Proof</u>: $c = q^3 - q^2 + 3q + 10$

Derivative of T is M

$\therefore \; TVC = q^3 - q^2 + 3q$

$TVC' = 3q^2 - 2q + 3$

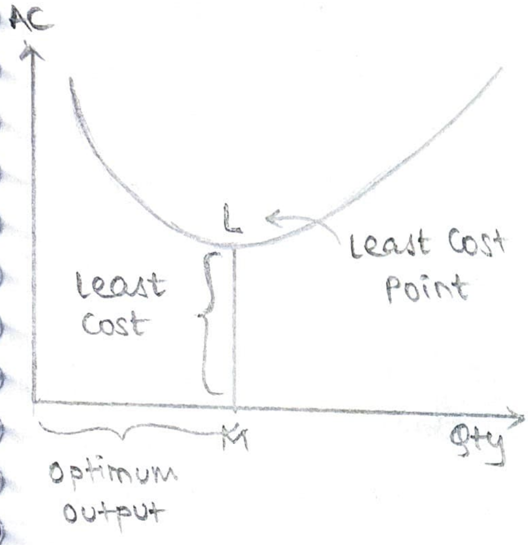
$MC = c' = 3q^2 - 2q + 3$

$\because$ TVC' = MC we conclude that TC is shaped by TVC.

- TFC is gap between TVC and TC.

Average Costs (AC)

- Total cost incurred per unit of the output produced.



Output produced at L is optimal output

$$AC = \frac{TC}{Q}$$

$$TC = TFC + TVC$$

$$\therefore AC = \frac{TFC + TVC}{Q}$$

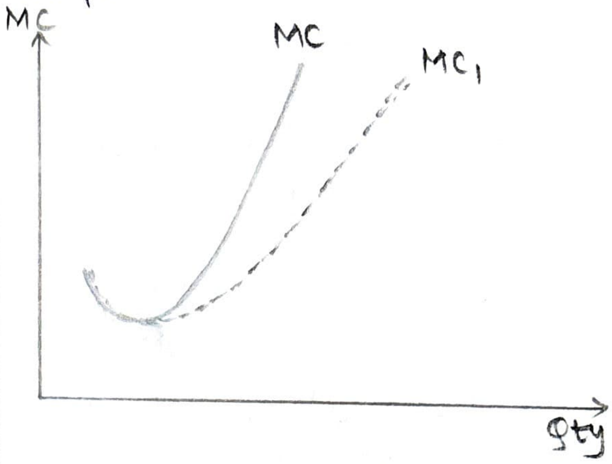
$$= \underbrace{\frac{TFC}{Q}} + \underbrace{\frac{TVC}{Q}}$$

$$AC = AFC + AVC$$

- AC curve is nearly 'U' shaped one

Marginal Cost (MC)

- Change in total cost incurred by producing an additional unit of output.



$$MC = \frac{d(TC)}{d(Q)}$$

- It's slopedness is indicative of the fixed factor efficiency

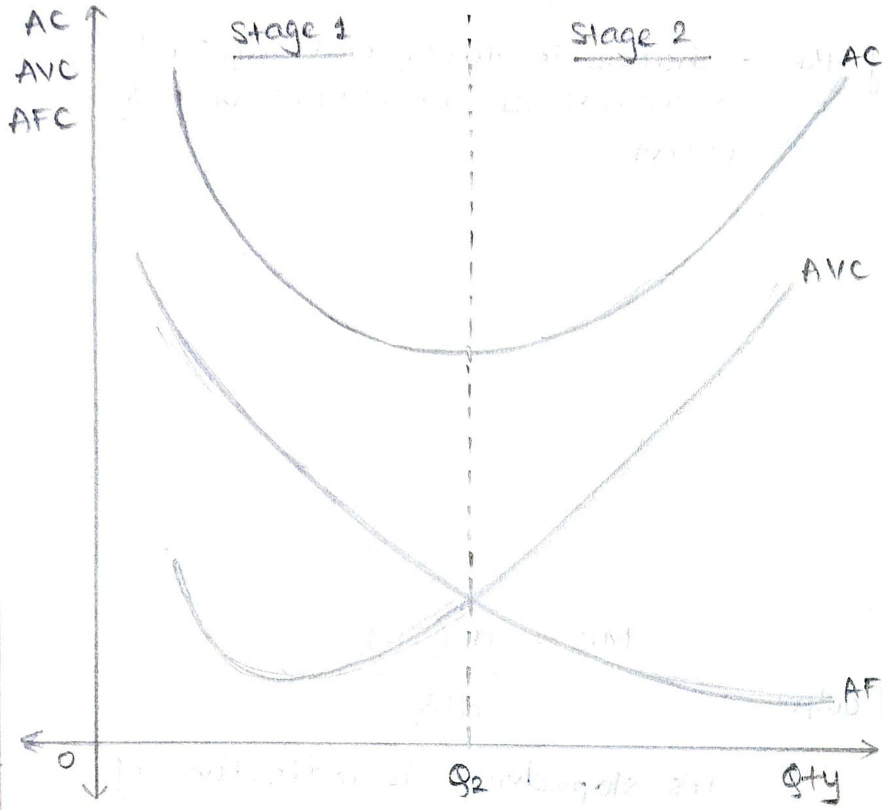
Efficiency of MC_1 > MC

- MC curve is nearly 'J' shaped one.

Deliberations on shape of the AC curve:

Q] Why is the AC curve 'U' shaped?

- AC is made up of AFC and AVC. Minima used as partition to get stage 1 and 2.
- In stage 1 AFC - continuously falling - it's positive. AVC is nearly zero. In vector algebra AFC is active ∴ it pulls the curve downwards.
- In stage 2, Fatigue is becoming exhaustive, if not replaced. variable cost increases very fast. AFC approaching zero / dying. Active vector is AVC and it pulls the curve towards the top.



- The nearly U shape of the AC curve can be explained as under (using AC's minima as a point of bifurcation into stage 1 and 2.

Stage 1 :

- The AFC continuously falls and this fall is financially robust (healthy) since fixed factor is mechanically fresh ∴ the firm enjoys internal economies of scale.
- The AVC falls and rises a bit because it is operating on mechanically fresh fixed factor. (Its taking full advantage of internal economies of scale)
- If we vectorize AFC and AVC, we conclude that $AC = AFC + AVC$

$$\underset{\downarrow}{AC} = \underset{\downarrow}{AFC} + \underset{\downarrow\&\uparrow}{AVC}$$

∴ ≈ zero

∴ In stage 1, AFC is the active factor hence in this stage AC, continuously falls.

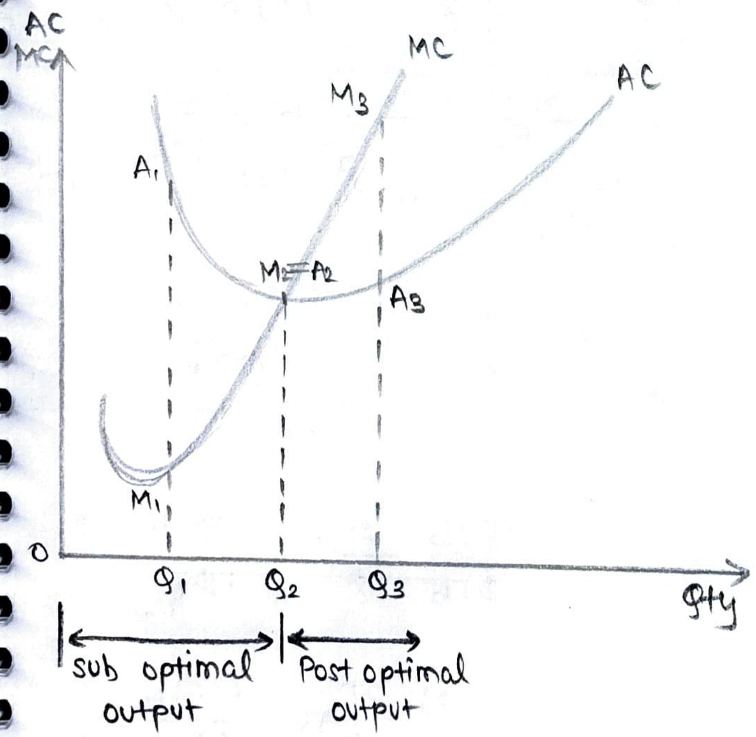
Stage 2 :

- The AFC continues to fall but this fall is financially not robust since Mechanical fatigue transforms to mechanical exhaustion giving rise to Internal diseconomies of scale.
- Since the variable factor is made to operate on fatiguing and exhausting fixed factor, the AVC continuously rises.
- If we vectorize AFC and AVC, the following can be inferred.

$$\underset{\uparrow}{AC} = \underset{\approx \text{zero}}{AFC} + \underset{\uparrow}{AVC}$$

∴ In this stage, the AC continuously rises. Hence the fixed factor and variable factor over the 2 stages causes the AC curve to be nearly a 'U' shaped curve. (nearly because whilst it rises, it experiences mechanical Fatigue and Mechanical Exhaustion)

Relationship between AC and MC



- It is possible to identify 3 definite relationships between AC and MC.

* At optimal Output :

- This occurs at AC's minima
- At this and only at this point, the AC and MC curves intersect.

∴ $MC(M_2, Q_2) = AC(A_2, Q_2)$

* Before / Sub optimal output :

- The AC continuously falls because of Mechanical freshness.
- The MC falls and rises because of Mechanical freshness and early stages of mechanical fatigue.
- MC lies below AC.

Eg: At output OQ_1, $MC(M_1, Q_1) < (A_1, Q_1)$

* <u>Post optimal output:</u>

- Both the AC and MC continuously rise
- The MC lies above AC

Eg: At output OQ_3, ₹ $MC(M_3, Q_3) > AC(A_3, Q_3)$

These relationships can be mathematically confirmed either through Geometry - Trigonometry method or Calculus Method.

- Geometry - Trigonometry Method:

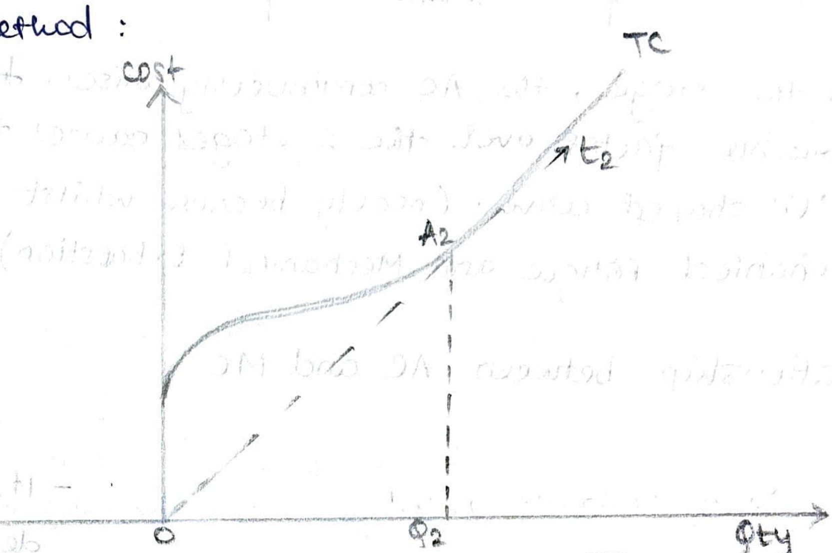
$\therefore$ at A_2

$MC = AC$

OQ_2 = Optimum Qty

$\tan \angle A_2 O Q_2$ = MC at OQ_2

= AC at OQ_2



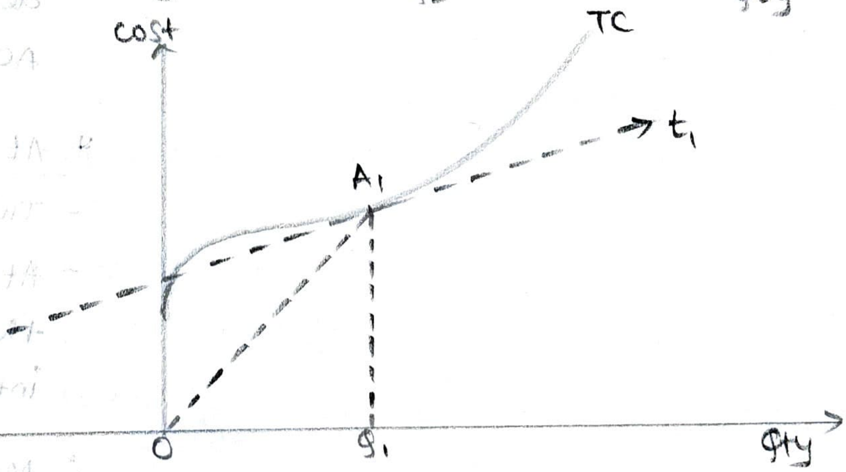
since at A_1,

$\tan \angle A_1 O Q_1 > \tan \angle A_1 T_1 Q_1$

AC > MC

$\tan \angle A_1 T_1 Q_1$ = MC at OQ_1

$\tan \angle A_1 O Q_1$ = AC at OQ_1



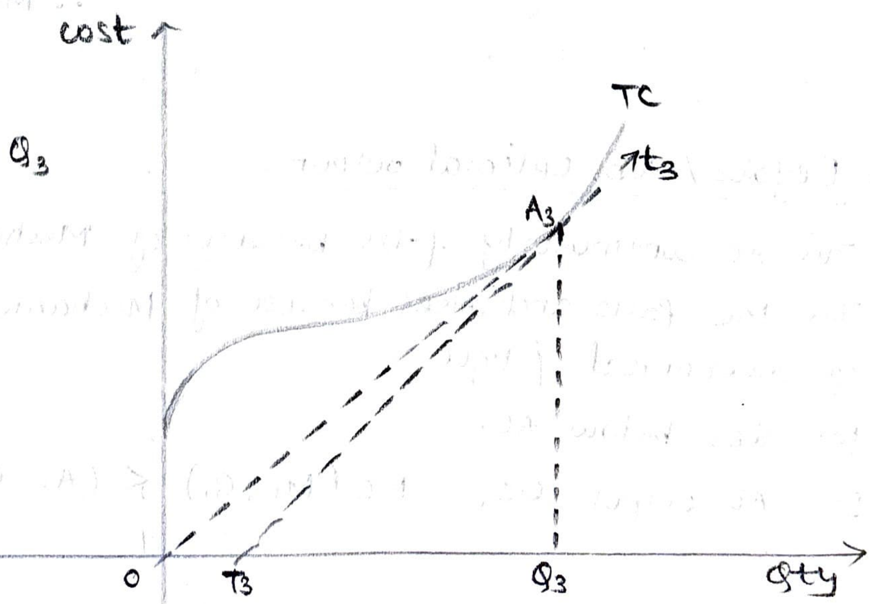
since at A_3,

$\tan \angle A_3 O Q_3 < \tan \angle A_3 T_3 Q_3$

AC < MC

$\tan \angle A_3 O Q_3$ = AC at OQ_3

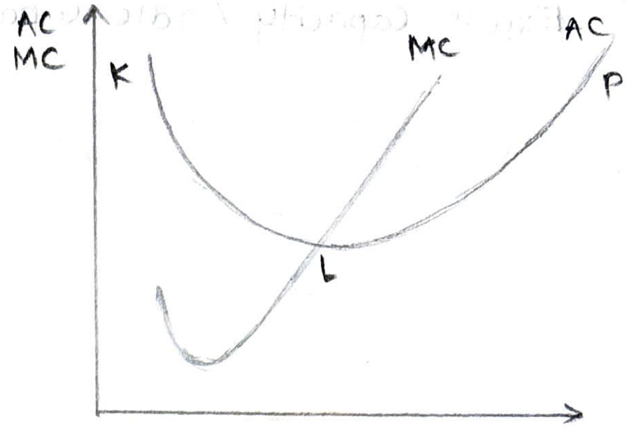
$\tan \angle A_3 T_3 Q_3$ = MC at OQ_3



- Calculus Method :

At Pre optimum output

$$AC' = \left(\frac{TC}{Q}\right)' < 0 \qquad \frac{d(TC)}{d(Q)} < 0$$



$$\therefore \frac{(Q)\dfrac{d(TC)}{d(Q)} - (TC)\dfrac{d(Q)}{d(Q)}}{Q^2} < 0$$

$$\frac{Q}{Q^2}\frac{d(TC)}{d(Q)} - \frac{TC}{Q^2}\frac{d(Q)}{d(Q)} < 0$$

$$\frac{1}{Q}\frac{d(TC)}{d(Q)} - \frac{TC}{Q^2}(1) < 0$$

$$\frac{1}{Q}\left(\frac{d(TC)}{d(Q)} - \frac{TC}{Q}\right) < 0$$

$$\frac{d(TC)}{d(Q)} - \frac{(TC)}{Q} < 0$$

$$\frac{d(TC)}{d(Q)} < \frac{TC}{Q}$$

$$MC < AC$$

At Post Optimum Output

$$AC' = \left(\frac{TC}{Q}\right)' > 0 \qquad \frac{d(TC)}{d(Q)} > 0$$

$$\frac{(Q)\dfrac{d(TC)}{d(Q)} - (TC)\dfrac{d(Q)}{d(Q)}}{Q^2} > 0$$

$$\frac{Q}{Q^2}\frac{d(TC)}{d(Q)} - \frac{TC}{Q^2}\frac{d(Q)}{d(Q)} > 0$$

$$\frac{1}{Q}\frac{d(TC)}{d(Q)} - \frac{TC}{Q^2}(1) > 0$$

$$\frac{1}{Q}\left(\frac{d(TC)}{d(Q)} - \frac{TC}{Q^2}\right) > 0$$

$$\therefore \frac{d(TC)}{d(Q)} - \frac{TC}{Q^2} > 0$$

$$\frac{d(TC)}{d(Q)} > \frac{TC}{Q} \qquad MC > AC$$

At optimum output

$$AC' = \left(\frac{TC}{Q}\right)' = 0$$

$$\frac{d(TC)}{d(Q)} = 0$$

$$\frac{Q\dfrac{d(TC)}{d(Q)} - TC\dfrac{d(Q)}{d(Q)}}{Q^2} = 0$$

$$\frac{Q}{Q^2}\frac{d(TC)}{d(Q)} - \frac{TC}{Q^2}(1) = 0$$

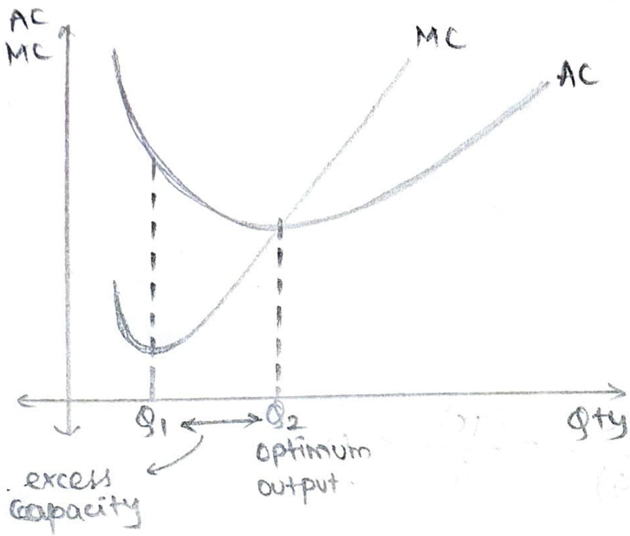
$$\frac{1}{Q}\frac{d(TC)}{d(Q)} - \frac{TC}{Q^2} = 0$$

$$\frac{1}{Q}\left(\frac{d(TC)}{d(Q)} - \frac{TC}{Q}\right) = 0$$

$$\frac{d(TC)}{d(Q)} = \frac{TC}{Q}$$

$$\therefore MC = AC$$

Excess Capacity / Idle Capacity :



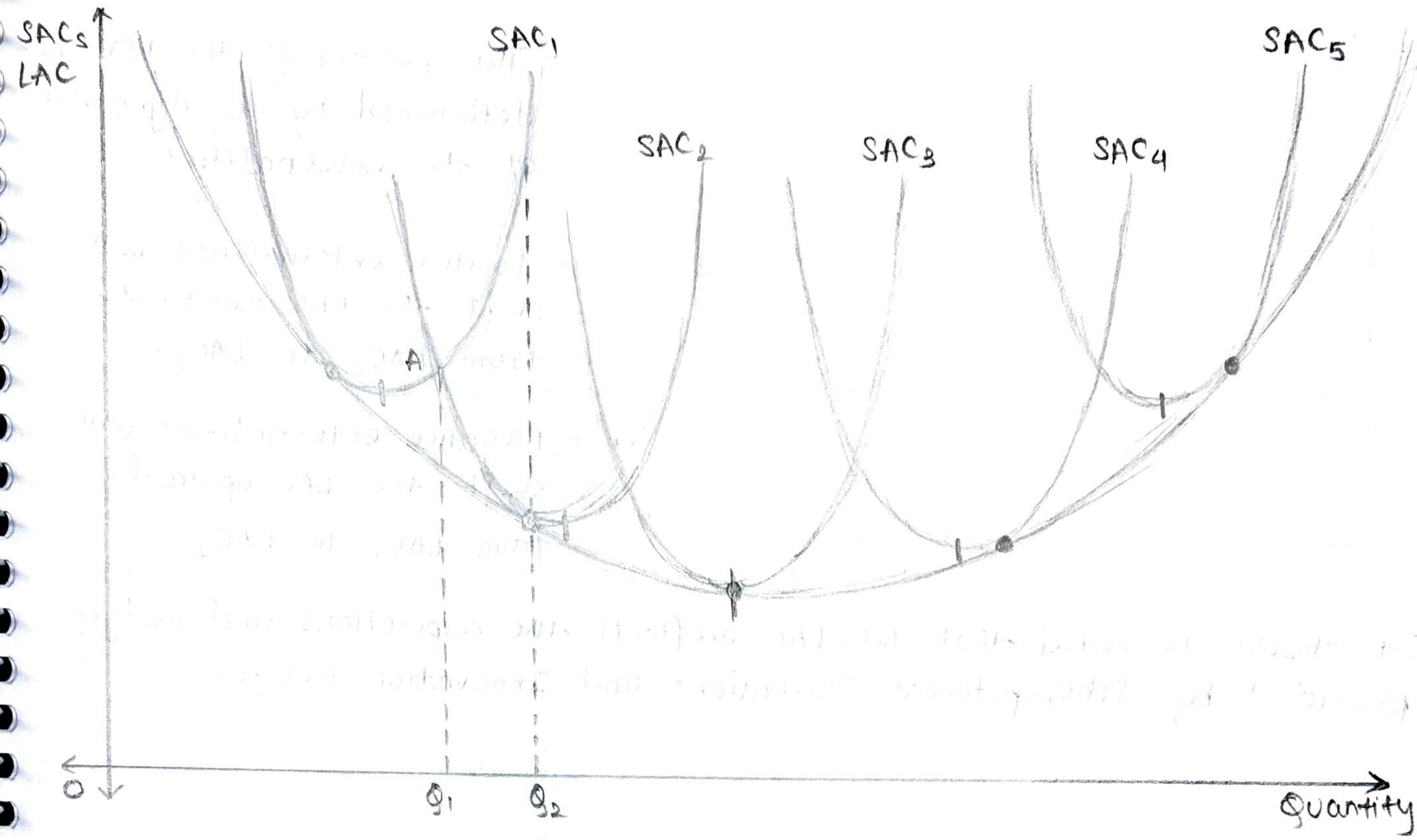
Excess capacity = $OQ_2 - OQ_1$
= optimum Qty − Actual Qty

Excess capacity can be interpreted as either :

Tragedy	Strategy
- It leads to building inflationary pressures since the set of firms would be operating with large excess capacity in place.	- Modern companies use excess capacity strategically.
- It aggrevates unemployment since less than optimal output is being produced, the number of resources that can be used reduces.	- They exhibit production agility - production is made responsive to market needs.
- It causes large scale resource wastage. The Raw material used to create the fixed factor is not optimally used.	- Seasonality compliance - this ensures inventories are minimal
	- It leads to scientific, inventory management with respect to both inputs and final products.

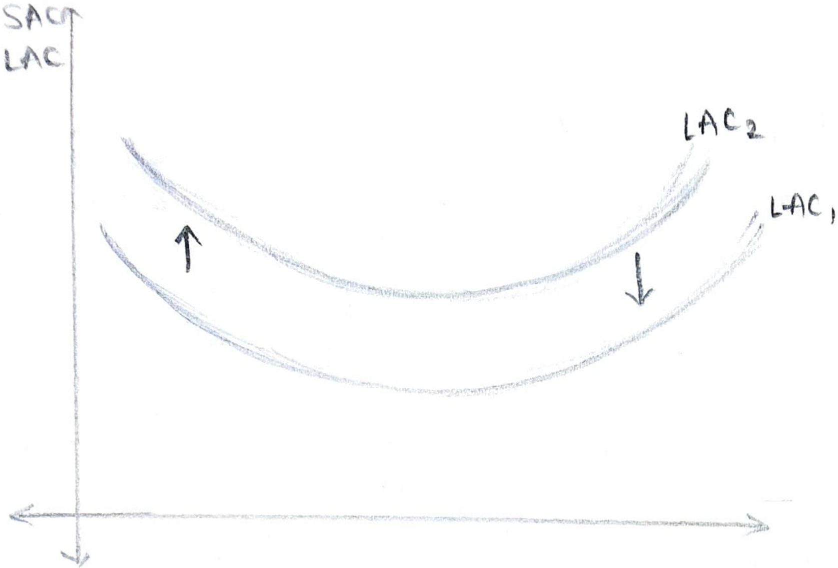
Long Run Cost Curve :

- In the long Run Fixed cost can be changed.
- One short Run Avg. cost curve represents one technology
- Every successive technology is more cost effective than the previous one.
- LAC is the locus of the operational points of every adopted technology
- LAC is envelope curve, planning curve.
- LAC is shaped by laws of Return to scale with IRS creating negative trajectory and DRS causing upward slope.
- Long Run Production cost influences the LAC.



- The LAC is the locus of all operational points of a technology that is available, accessible, affordable, adoptable and adaptable.
- The LAC is also called Envelope curve since it contains the Map of technologies i.e. available, accessible, affordable, adaptable, adoptable.
- The LAC is also called Planning curve due to foll. deliberations :
  - Point A is concurrent point between technologies exhibited by SAC_1 and SAC_2, whichever technology is adopted, the AC will be AQ_1 with an output OQ_1. Suppose both firms decide to produce Q_1, Q_2 more.
  - The firm operating from SAC_1 will have an AC as high as A_1Q_2 (so its price will have to be higher) But the firm operating with SAC_2 will experience an AC as low as A_2Q_2 (so its price will be lesser than viable firms). The firm using SAC_1 technology would tend to carry an inventory of Q_1Q_2 and other firm will not have this burden. Thus, the LAC provides firms with a Nudge as to the scheduling of its tech migration.
- The LAC is shaped by the operations of Laws of Returns to scale; IRS will shape its downward trajectory and DRS will shape the upward trajectory.

NOTE: DRS emerges when a ~~very~~ superior technology is being made operational when the appropriate and adequate support systems (infra) are not in place.



- The position of the LAC is determined by the dynamics of the externalities.

- Positive externalities will shift the LAC downwards from LAC_2 to LAC_1.

- Negative externalities will shift the LAC upwards from LAC_1 to LAC_2

It should be noted that the LAC reflects the aspirations and nudges provided by Schumpeterian Inventions and Innovation nudges.

REVENUE ANALYSIS

- Revenue and only revenue reveals the competitive nature of the market.
- Under perfect competition AR = Price.

Perfect Competition | Imperfect Competition

Schedules

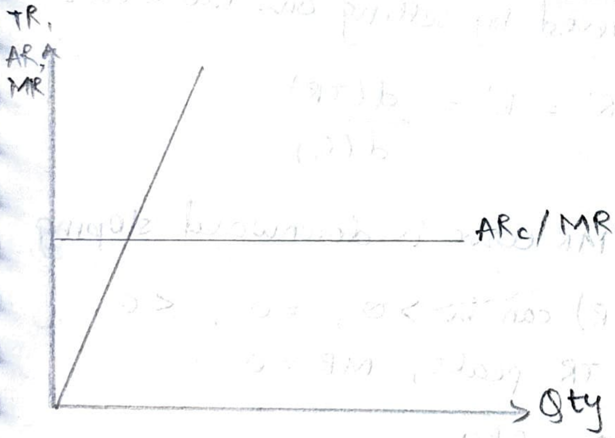
Perfect Competition

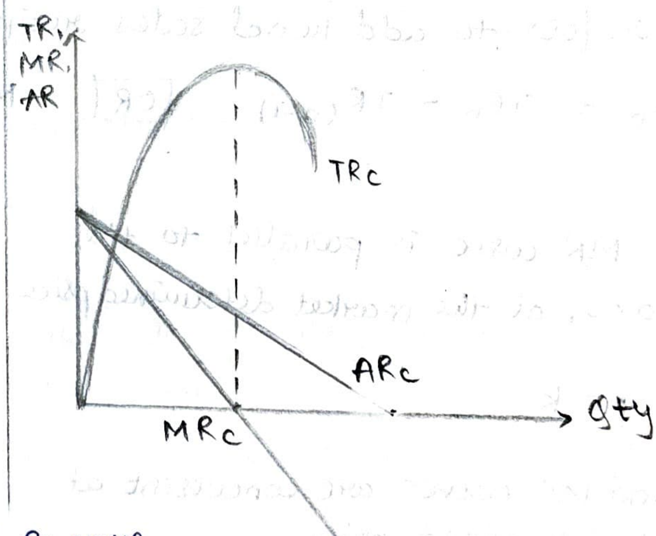
Qty	AR	TR	MR
1	2	2	2
2	2	4	2
3	2	6	2
4	2	8	2
5	2	10	2
6	2	12	2
7	2	14	2

Imperfect Competition

Qty	AR	TR	MR
1	30	30	30
2	27	54	24
3	23	69	15
4	20	80	11
5	12	60	-20
6	6	36	-24
7	1	7	-29

Graphical Representation





Total Revenue

- TR refers to the overall sales receipts recieved at given level of sales

$$TR = AR \times Q$$

$$TR = \int_0^Q MR\,(dQ)$$

- The TR curve :
1) is unishaped
2) Rises continously

Average Revenue

- AR refers to sales reciepts per unit of sales.

$$AR = \frac{TR}{Q}$$

* AR curve is the demand curve

AR is price.

Proof:

$TR = (AR)(Q)$ and $T.\ Exp. = (P)(Q)$

But, $TR = TE$ therefore,

$AR\ \cancel{(Q)} = P\ \cancel{(Q)}$

$\therefore AR = P$

Marginal Revenue

- MR refers to additional sales recipts recieved by selling one more unit.

$MR_n = TR_n - TR_{(n-1)}$ $\boxed{OR}$ $MR = TR' = R' = \dfrac{d(TR)}{d(Q)}$

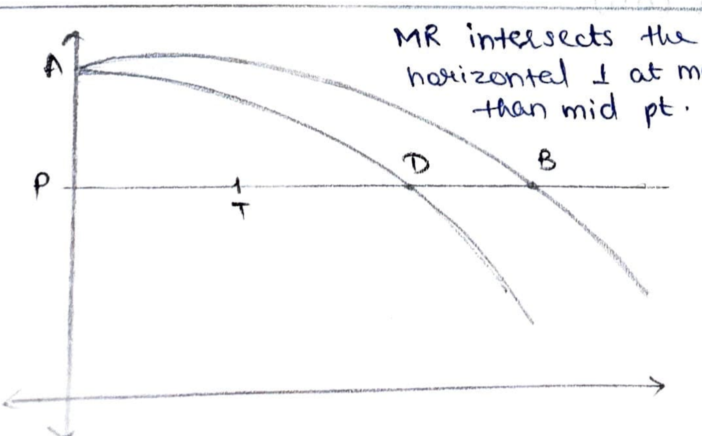
- The MR curve is parallel to the Qty axis, at the market determined price
- $R' = p = k$
- AR and MR curves are concurrent at market determined price.

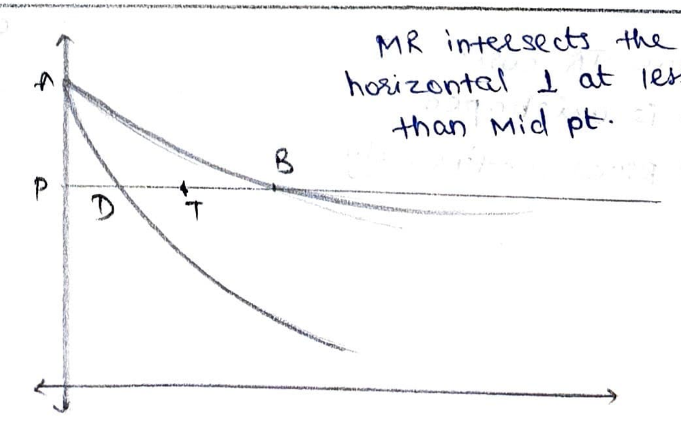
Such Firms can opt for profit maximization or breakeven status but they cannot opt for sales maximization.

- The MR curve is downward sloping
- $v(MR)$ can be > 0, $= 0$, < 0

when TR peaks, $MR = 0$

- $R' = a - 2kq$
- The objectives of such firms can be profit maximization, breakeven and even sales maximization.





Q] If the AR function is $p = 43 - 0.2q$ then prove that this data is from a firm operating under imperfect competition.

→ The $TR = AR(Q)$

$= 43q - 0.2q^2$

$MR = R' = 43 - 0.4q$

- This firm operates under imperfect competition since the TR function is quadratic and not linear.
- Slope of demand curve = 0.2 and not 0
- The slope of MR curve is 2 times the slope of the MR curve and the slopes are not equal.

Price at 20 units of sales is $P_2 = 43 - 0.2(20) = 39$

- Prices at different quantities are not uniform, this firm is operating under imperfect competition.

Q] If Price of product = ₹30 ($P = 30$) confirm that this data is from perfect competition.

AR : $P = 30$

∴ TR : $R = 30q$

MR : $R' = 30$

- Since AR = MR, the firm operates under perfect competition
- The prices are uniform.
- Since TR function is $R = 30q$, the corresponding curve will be linear.
- The slope of AR and MR curves = 0

Since slope of MR = 2 (slope of AR), the following commercial inferences can be drawn :

- It shows that TR changes at twice the rate of change in the price
- An increase in price would cause the TR to increase at twice that rate.
- It shows that a decrease in price would indicate that TR decreases at two times that Rate.

Relationship b/w AR, MR and e_{dp}

$$\boxed{e_{dp} = \frac{AR}{AR - MR}}$$

OBJECTIVES OF THE FIRM

1] Profit Maximization (Profit = π)

*** Marginal Method :**

Total Profit = TR − TC

π = TR − TC

To maximize π, $\pi' = 0$

$\therefore (R-C)' = 0$

$R' - C' = 0$

$\therefore MR - MC = 0$

$\therefore MR = MC$ → Necessary condition

However, the necessary condition creates 2 equilibrium points E_1 (OQ_1 units) and E_2 (OQ_2 units)

$\therefore$ In order to confirm which output would actually lead to maximum π, we use the second order condition for maxima.

$\pi'' < 0$

$(R-C)'' < 0$

$(R' - C')' < 0$

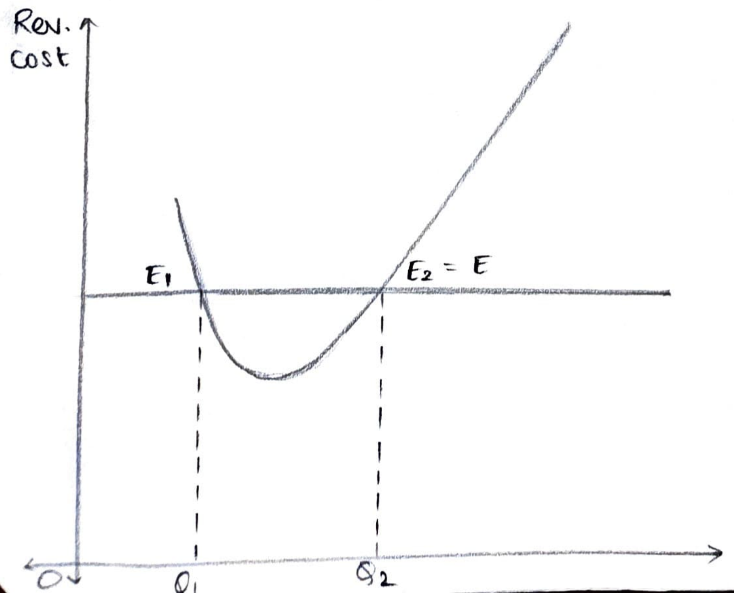
$(MR - MC)' < 0$

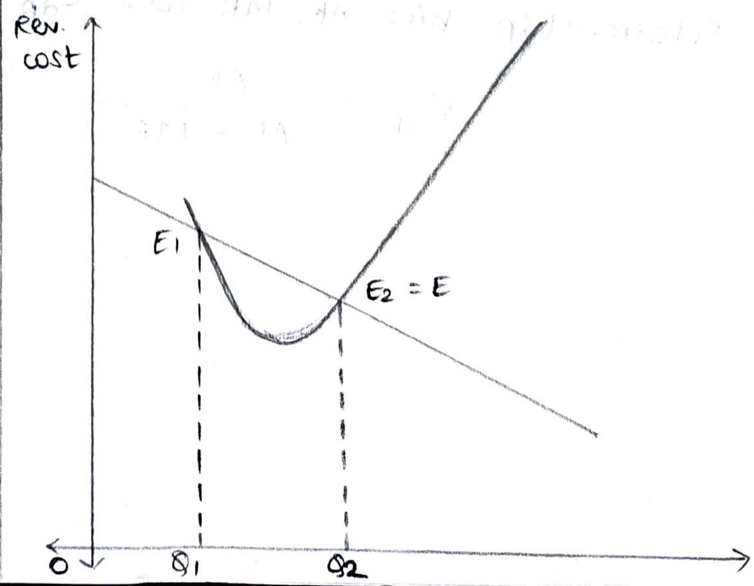
$MR' - MC' < 0$

$MR' < MC'$

$\therefore$ Slope of MC > Slope of MR → sufficient condition.

Perfect Competition	Imperfect Competition

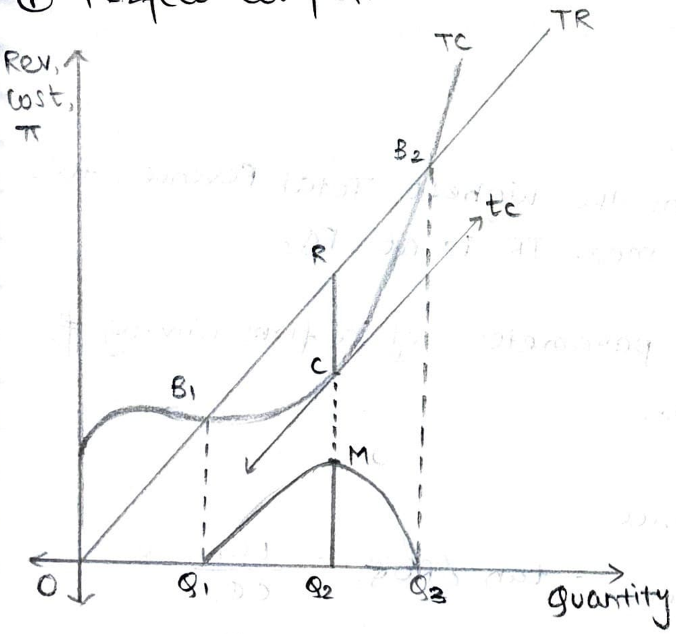




- At E_1, the slope of MC is lesser than the slope MR
- The necessary condition is fulfilled but however the sufficient condition is not fulfilled ∴ Equilibrium E_1 (output : OQ_1) is rejected.
- At E_2, slope of MC > Slope of MR since both the necessary and sufficient conditions are fulfilled, this firm will maximize profit at only output OQ_2.
- E_2 is stable equilibrium point because :
  - producing a little less than OQ_2 would mean that the firm sacrifices the maximization of profits ∴ such a move is not an incentivized one.
  - If the firm produces a little more than OQ_2 would ~~mean~~ ~~it will~~ begin to incur losses hence this too is not an incentivized move.
- Since the firm has no incentives to shift to the LHS / RHS of OQ_2, equilibrium point E_2 is a stable equilibrium point.

* Total Method :

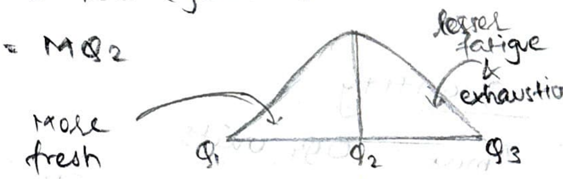
① Perfect Competition -



- At output OQ_2 and only output OQ_2 profits (π) are maxima
- The value of maximum $\pi = \frac{RC}{MQ_2}$

$Price_{B_1} = Price_{B_2} = Price_R = \tan \angle B_2OQ_3$

$TR = RQ_2$
$- TC = CQ_2$
Profit = RC (greatest)
$RC = MQ_2$

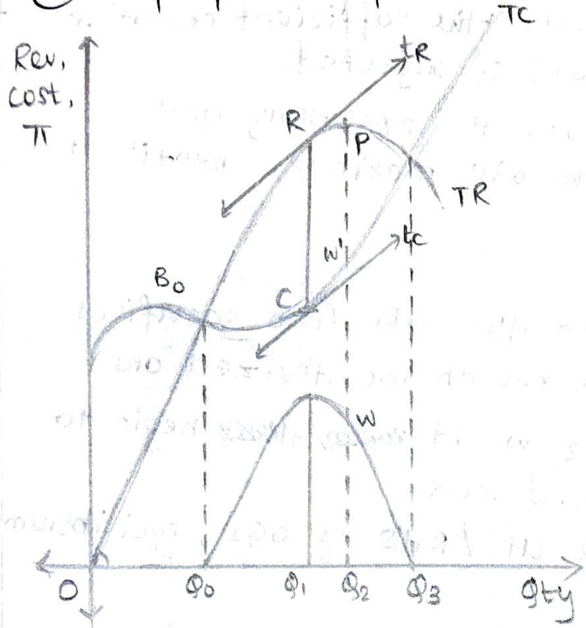


Q] Show profit maximization under perfect competition using Total Method. Hence, prove prices are uniform under PC $\boxed{OR}$ under PC, the law of Demand does not operate.

?

→

(ii) Imperfect Competition :



Value (π Max.) $= RC = MQ_1$

Q (π Max.) $= OQ_1$

Price B_0 = tan $\angle B_0 O Q_0$

Price B_2 = tan $\angle B_2 O Q_2$

Price R_2 = tan $\angle R O Q_1$

Q] Prove that prices are not uniform under imperfect comp. [OR] Laws of Demand operates under imperfect competition.

2] Sales Maximization

- Sales $_{max}$ is not quantitative variable
- Sales is maximized when the firm earns the highest Total Revenue, this occurs at OQ_2 and the corresponding max. TR is at PQ_2

Q] Compare and contrast the commercial parameters of a firm aiming for π_{max} or breakeven statuses or sales $_{max}$.

Quantity :

π_{max} = OQ_1 units

Sales $_{max}$ firm = OQ_2 units

BE_1 firm = OQ_0 units

BE_2 firm = OQ_3 units.

Price :

$\pi_{max.}$ = tan $\angle ROQ_1 = \frac{RQ_1}{OQ_1}$

Sales $_{max.}$ = tan $\angle POQ_2 = \frac{PQ_2}{OQ_2}$

BE_1 = tan $\angle B_0 O Q_0 = \frac{B_0 Q_0}{OQ_0}$

BE_2 = tan $\angle B_2 O Q_3 = \frac{B_3 Q_3}{OQ_3}$

Profits :

$\pi_{max.} = RC = MQ_1$

Sales $_{max.}$ = $PW' = WQ_2$

$BE_1 = BE_2$ = zero ($\because$ TR = TC)

- For sales $_{max}$, $TR' = 0$

$\Downarrow$

$MR = 0$

Q] Can a sales max firm be a BE firm
→ Yes, when TC intersects TR at it's (TR's) peak.

Q] Can a sales max firm be a loss making firm
Eg: - Printer low cost - cartridge high cost
- PS 5 low cost - Game CPs higher cost
- Public sector transport BEST buses - low cost initially Good AC buses.
- The telecom service rates adopted this motivation of making mobile phone a public good.

- Sales maximization is possible only under imperfect competition and not under perfect competition because in the former case, the TR function can be maximized. In the later TR does not have a maximum position.

TR : $R = 25q - 0.2q^2$ $\qquad$ $C = 5q^3 - 2q^2 + 7q + 100$

From the above data do as directed:

Rough Work:

$AR = 25 - 0.2q^2$ $\qquad$ $AC = 5q^2 - 2q + 7 + \frac{100}{q}$

$MR = 25 - 0.4q$ $\qquad$ $MC = 15q^2 - 4q + 7$

$\pi(fn)$: $\pi = TR - TC$

$\pi = 25q - 0.2q^2 - 5q^3 + 2q^2 - 7q - 100$

$= -5q^3 + 1.8q^2 + 18q - 100$

Q] Find : the output that the firm would produce if it followed the max principle.

If the firm adopted the $\pi_{max.}$ principle then

$\pi' = 0$

$\pi' = -15q^2 + 3.6q + 18 = 0$

$q_1 = 1.22$, $q_2 = -0.98$ and accept only $q_1 = 1.22$

∴ For this firm, stable equilibrium is at $q = 1.22$

Q] Find the profit status of such a firm,

V_π at $q = 1.22$ is $-5(1.22)^3 + 1.8(1.22)^2 + 18(1.22) - 100$

$= -₹84.2$

∴ It would be advisable for this firm to avoid subnormal (negative) profits by not following the $\pi_{max.}$ principles.

Q] Suppose this firm continues producing $q = 1.22$ units, at what price would it sell?

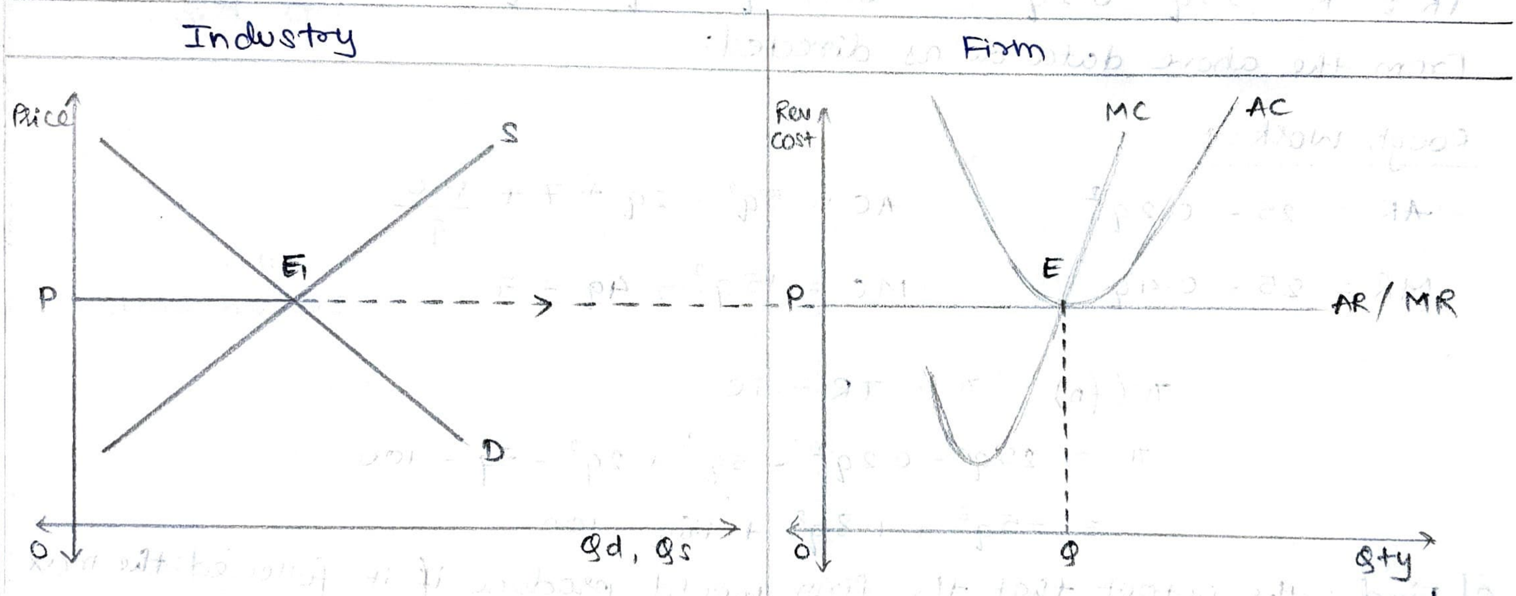
$$AR = p = 25 - 0.2(1.22)$$
$$= 25 - 0.244 = ₹\ 24.76$$

Q] If this firm opts for sales max what should be the output it would have to sell?

$$MR = 0$$
$$0 = 25 - 0.4(q)$$
$$0.4q = 25 \qquad q = \frac{250}{4} = 62.5 \text{ units.}$$
$$AR = ₹\ 12.50$$

SHORT RUN DYNAMICS UNDER PERFECT COMPETITION

It is possible for such a firm to either enjoy super normal profits ($\pi > 0$) or subnormal profits ($\pi < 0$) or Normal profits ($\pi = 0$)



Area ($\square$ OPEQ) = TR
Area ($\square$ OPEQ) = TC

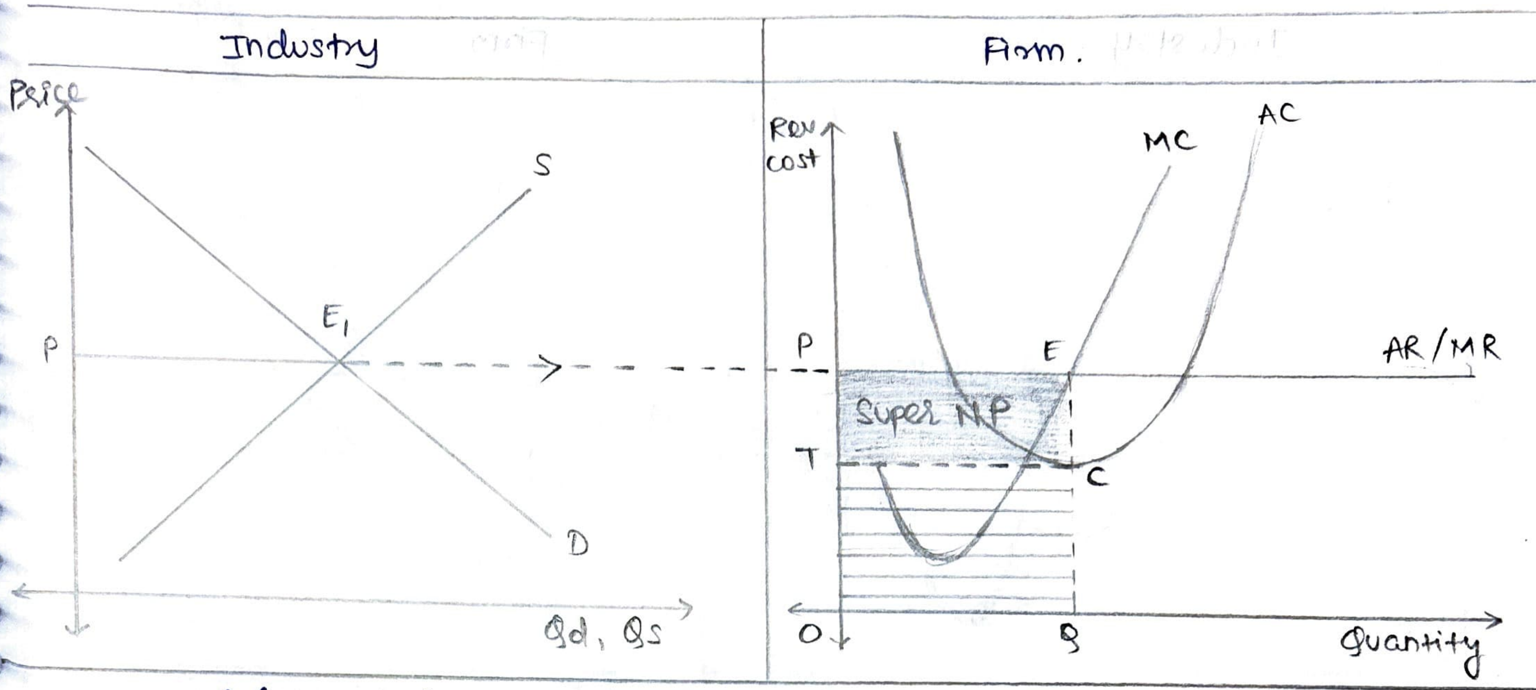
$T(\pi)$ = zero as TR = TC

Case 1:

At point E the firm reaches stable equilibrium since MC = MR and the slope of MC > Slope of MR ∴ this firm at price OP produces OQ units However since TR [A ($\square$ OPEC)] = TR = TC [A ($\square$ OPEC) ∴ π = zero] and the firms performs as a breakeven firm.
Many NGOs and charitable institutions perform as such not for profit firms

Q] Discuss Market Mechanisms under perfect competition showing π +ve / $\pi = 0$ / π -ve (continue) v/s π (shutdown).

Supernormal Profit



A (□OPEQ) = TR

A (□OTCQ) = TC

A (□ TPEC) = +ve T(π) as TR > TC

~~At~~ stable equilibrium, point E :

MC = MR = AR > AC

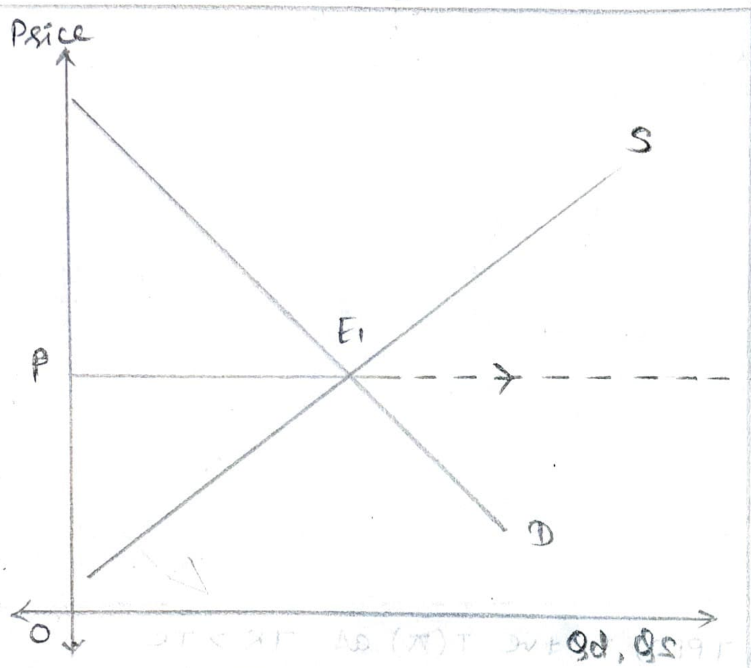
In this situation, the firm's total expenditure (at stable equilibrium) is A (□OPEQ) but with Market prices being OP, which is more than OT by PT, the firms overall supernormal profit = A (□OPEQ).

These firms are the usual commercial entities that ensure that their revenues more than cover costs.

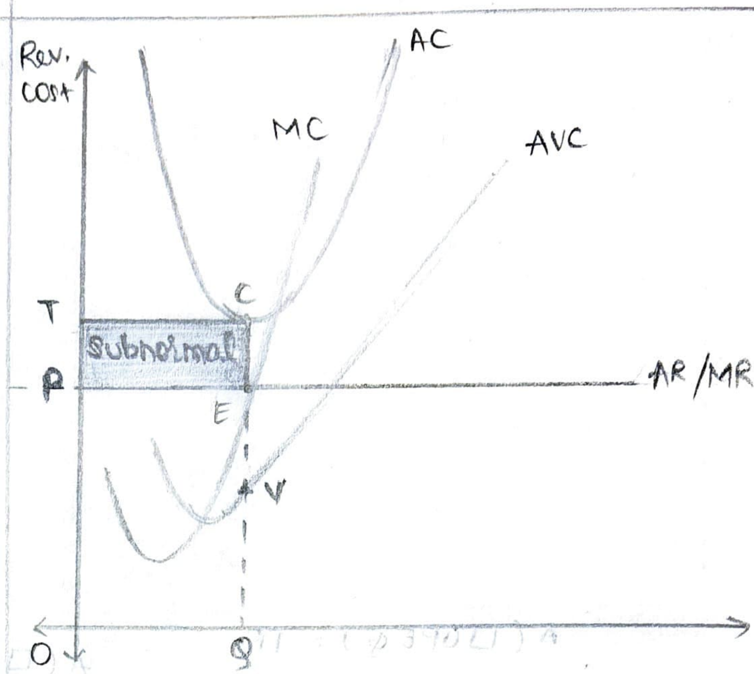
The supernormal profit in an economy like India would be subject to deductions like CIT (corp. income tax) - 22% and such firms are obliged to subscribe 2% of total π, CSR. If this Joint stock company, a part of post deducted π has to be shared among the shareholders as dividend and post that upto 40% of the profit can be invested in speculation and remaining has to be ploughed back into the company.

Subnormal Profits (continue) - 3 (a)

Industry



Firm



A (□OPEQ) = TR

A (□OTCQ) = TC

A(□TPEC) = -ve T(π) as TR < TC

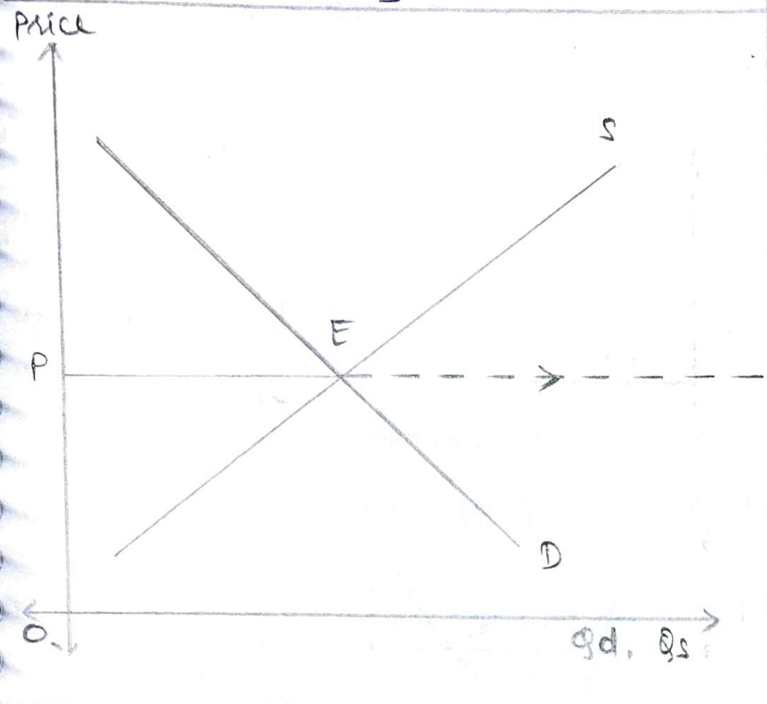
At stable equilibrium point E,

MC = MR = AC < C , AR > AVC

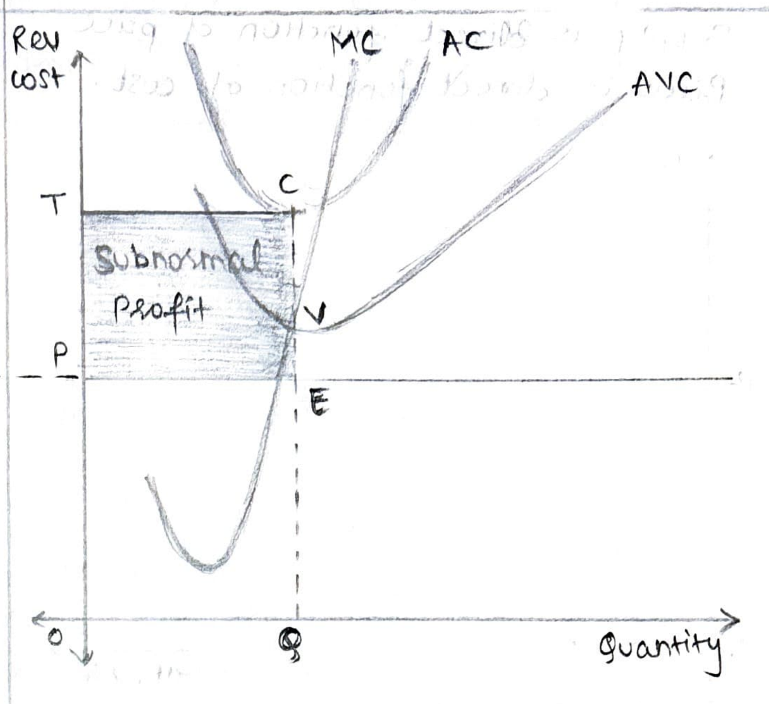
In this scenario, the total cost (at stable eq.) is A(□OTCQ) = TC, which is greater than the Total Revenue [A(□OPEQ) = TR] earned by this firm ∴ the subnormal profits = A(□PTCE). Since this firm is able to cover its operational expenditures through the price. (Price EQ > Variable cost VQ) by VE since the firm fully covers opex and partly covers capex. This firm would be advised to continue production for some more time hoping that when other firms shut down, the revenue situation for this firm improves. It is financially sick company and such firm will not be allowed to file for bankruptcy.

Subnormal Profits (shutdown) - 3 (b)

Industry



Firm



A (□OPEQ) = TR

A (□OTCQ) = TC

A (□TPEC) = -ve T(π) as TR < TC

AR < AVC

In this situation the concerned firm's total cost is as high as OTCQ and revenue is as low as A(□OPEQ). The huge subnormal profit A(□TPEC) would compel this firm to shut down. Financially, such firms are councelled to shut down since their AVC (QV) is more than the price QE such firms would be legally allowed to file for bankrupcy.

AC = AVC + AFC

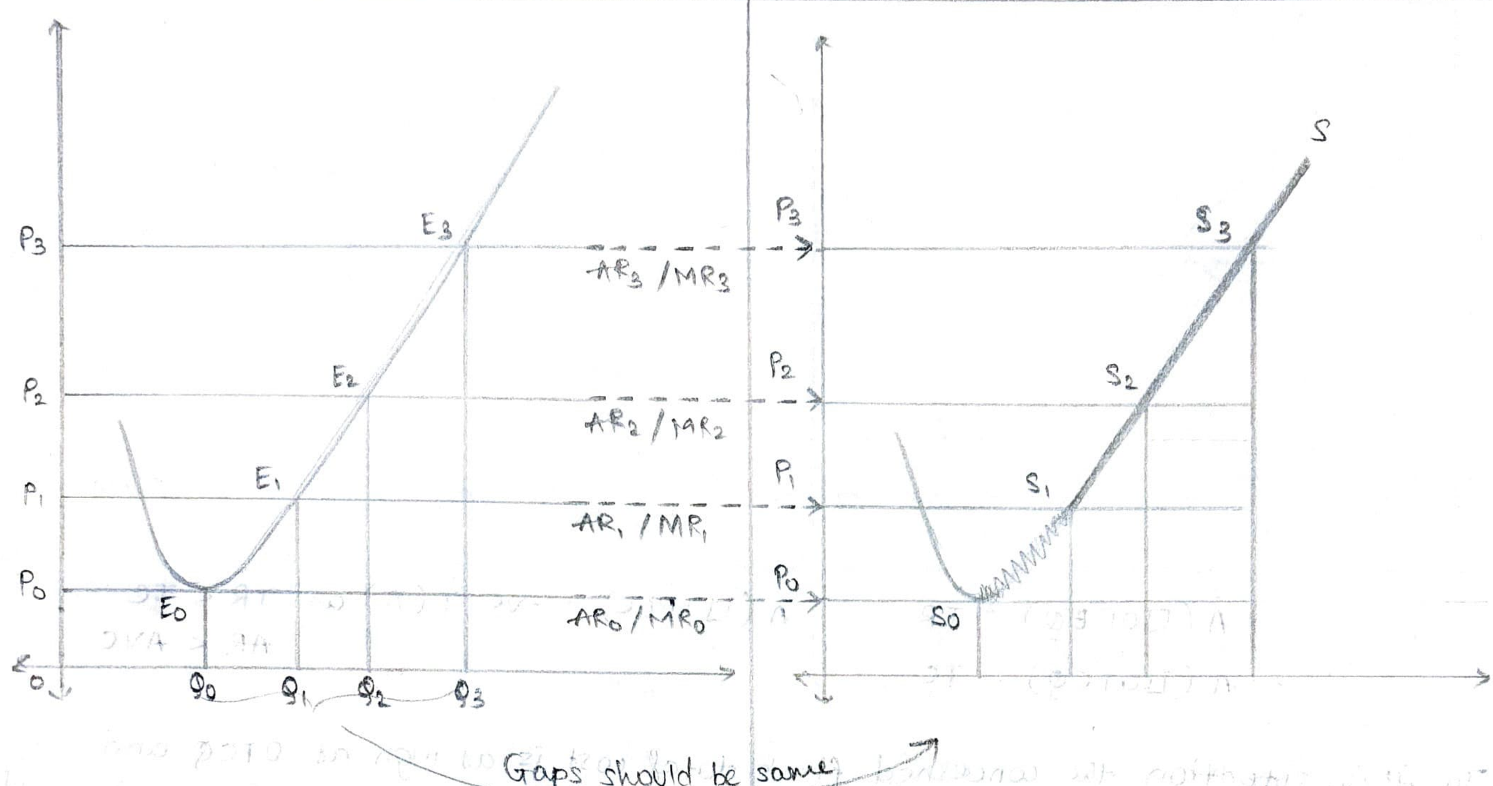
CQ = CV + VQ

$$CQ = \underbrace{CV}_{\text{fully uncovered}} + (\underbrace{VE}_{\text{uncovered}} + \underbrace{EQ}_{\text{covered}})$$

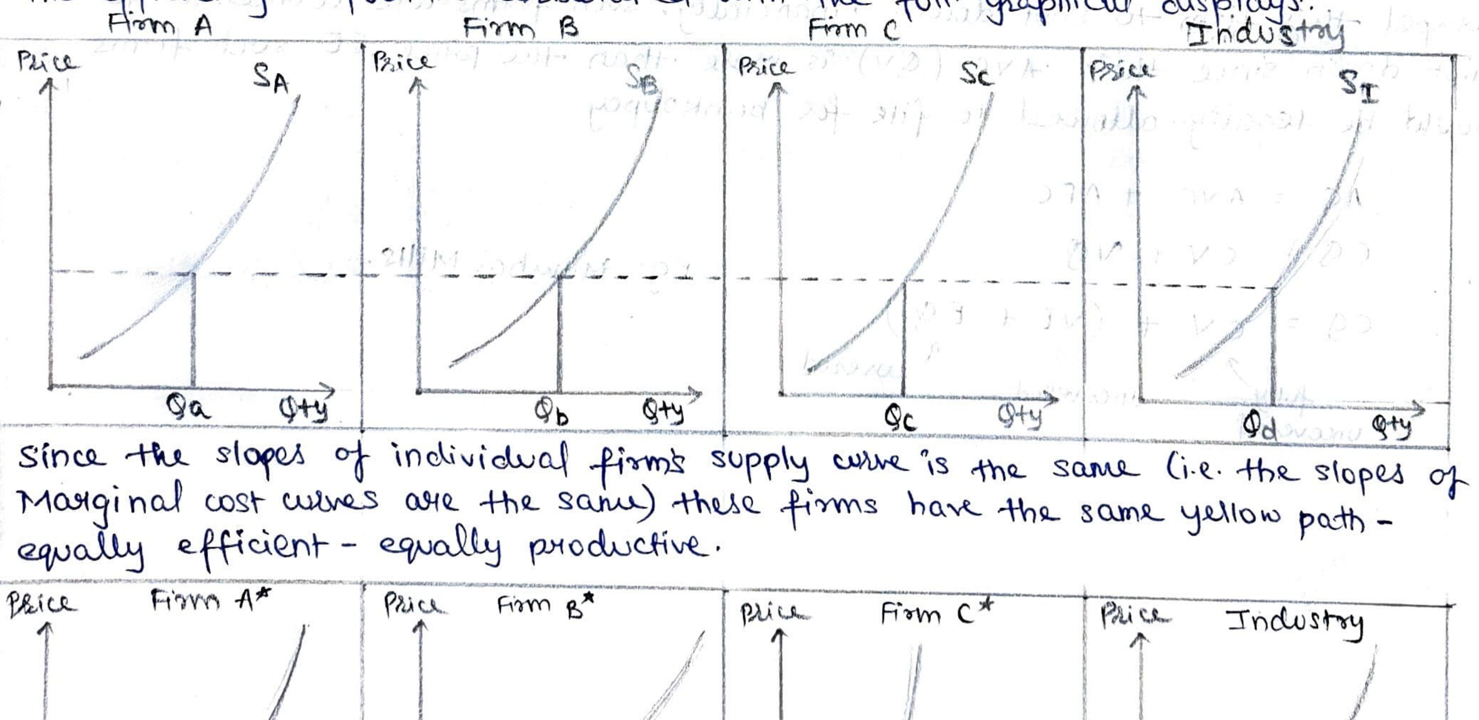
Eg: Mumbai Mills.

PRODUCTION EFFICIENCY UNDER PERFECT COMPETITION

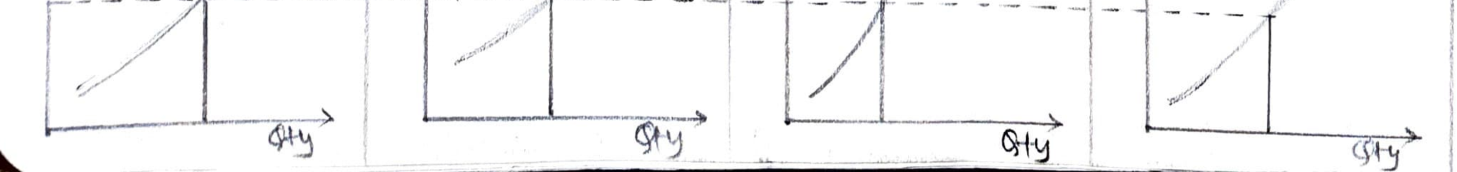
- The upward sloping MC curve above AVC is the supply curve of the firm.
  Supply is direct function of price
  Price is direct function of cost.



The efficiency is further corborated with the foll. graphical displays.



Since the slopes of individual firm's supply curve is the same (i.e. the slopes of Marginal cost curves are the same) these firms have the same yellow path - equally efficient - equally productive.



Since the Slopes of firms involved in this industry differ the Marginal costs slopes also differ.
- Steeper the slope of MC (firm C^*), less efficient is that firm.
- Gentler the slopedness of MC curve (firm B^*) more efficient is that firm

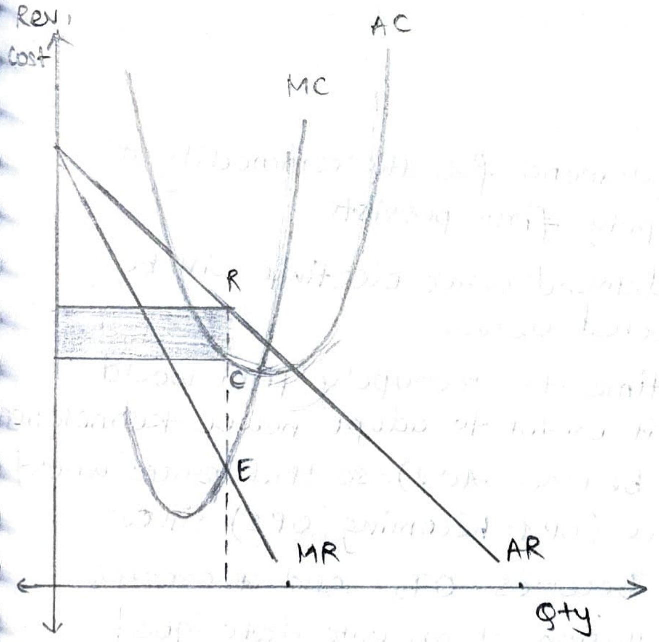
NOTE : The summation of each firm's supply is indicative of industry's supply situation.

MONOPOLY

A monopoly firm is a single seller situation in which there are no close substitutes (TINA)

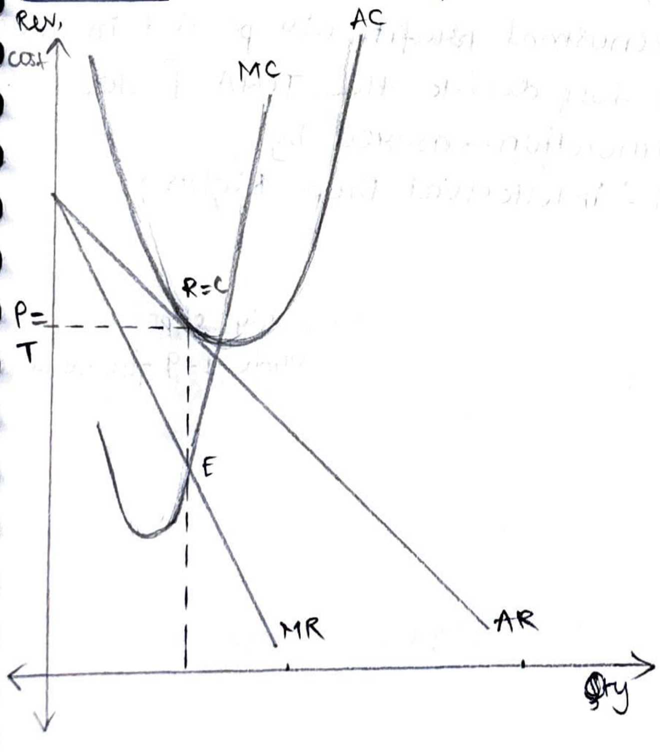
Operations of a Monopolyst firm under different profit conditions.

Case 1: Supernormal Profit (short run)



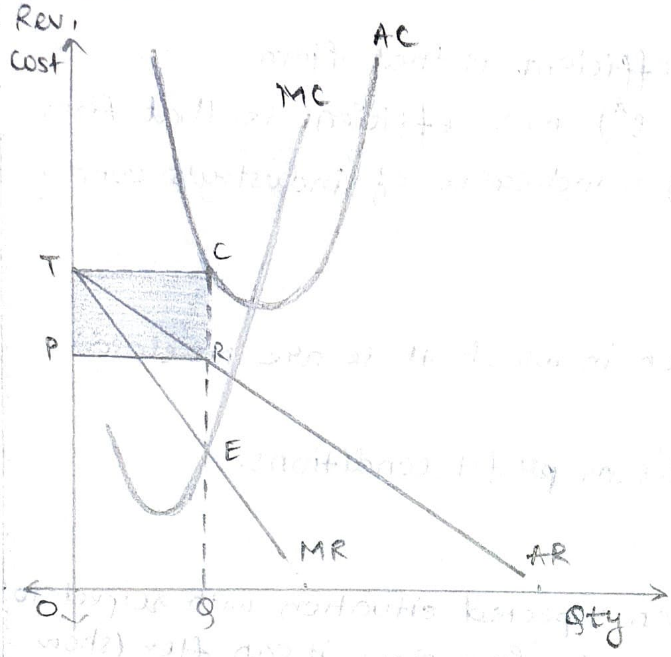
- It is an expected situation with respect to the monopoly firm since it can flex (show off) its commercial power especially when it has the TINA factor on its side. This normally happens in urban areas, where, conspicuous (demonstrative) consumption takes place.
- It is in urban spaces that finance is the most ubiquitous.
- Due to the Duesenberry (demo) effect and Ratchet effect (inability of a consumer to adopt to a lesser standard due to a fall in the income)

Case 2: Normal Profit (Short Run)



- Peri urban space, the space where consumption can be impacted through demonstration effect.
- The monopoly firm, to increase it's market penetration engages with this segment of the market hoping, at some later date, this market segment will convert this novelty consumption into necessary consumption.
- By posing as a charitable firm the monopoly firm is able to mitigate the negative aspect of the monopoly firms.

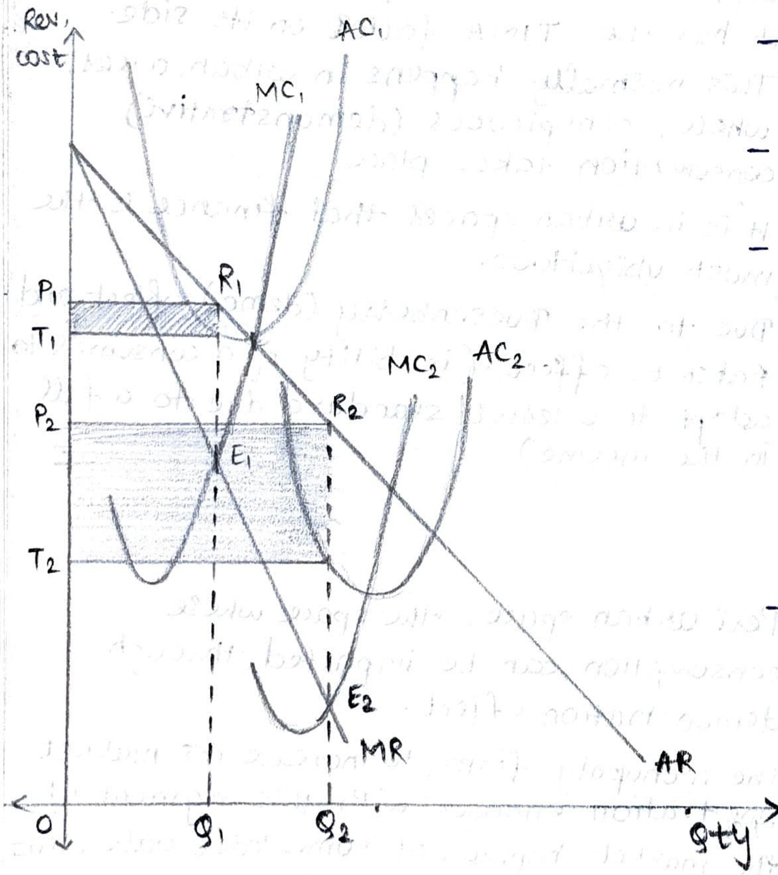
Case 3: Sub-Normal Profits (Short Run)



- The monopoly firm, to get the first mover advantage, engaging with this low purchasing power segment of the society in anticipation that at some point of time, the consumption basket would normalize the consumption of this good.
- It helps the company in gaining social recognition.

Q] Explain monopoly firm's operation in the short Run 1/2/3 ?

Long Run :



- The demand for the commodity of monopoly firm persists.
- The demand curve overtime will be inflected curve.
- Overtime the monopoly firm would find it useful to adopt newer technologies (AC 1 becomes AC 2) so that prices would reduce (OP 1 becoming OP 2) since OT_1 becomes OT_2. and it creates the illusion of an upto date 'good' being sold by the monopoly firm.
- Supernormal profits will persist in the long run due to the TINA factor continuation. ensured by IPR - intellectual Prop. Rights.

Monopoly chpt: Slide 1-9 for exam

MONOPOLISTIC COMPETITION

Monopolistic competition was coined by E.H. Chamberlin (USA, 1939)

Brand v/s Brand v/s Brand.

- The death of monopoly due to entry of competition in the Market.
- Imperfect comp. - Joan Robinson (UK, 1939)

Q] Discuss the features of Monopolistic Competition:

The following are the salient features of Monopolistic Competition: Intro for the ans.

1) The Collective Nature of the firms involved -

a) According to Chamberlin, the Range of no. of firms involved in such a market format is from 20 - 100 (formal / informal sector) i.e. how he saw that many firms are involved in the production of a good that satisfies the same want.

b) For Chamberlin the no. of firms that compose the term 'Many' is not large enough to be called an industry. Hence Chamberlin preffered the term 'Group' to describe the term collection of firms for particular product.

c) By using the term 'Group', Chamberlin points out to 'Interdependence among sellers' this leads to vibrant rivalry among the existing firms, giving rise to the Schumpeterian Dynamics. This means that entry / exit of firms affects the Dynamics wrt the product being discussed.

d) This gives rise to different firms opting for various objectives like sales max, Profit max, satisfying firm, welfare objective. Different objectives would require different pricing strategies in order to satisfy the market.

* Satisfising - satisfied with return and will survive.

2) The Nature of Commodity involved -

a) Chamberlin noticed that firms indulge in product differentiation i.e. cosmetically (aroma, colour, size) the product exhibited differences but functionally they serve the same purpose. Furthermore, different versions (upper and lower end) are created to cater to different income segment in the market.

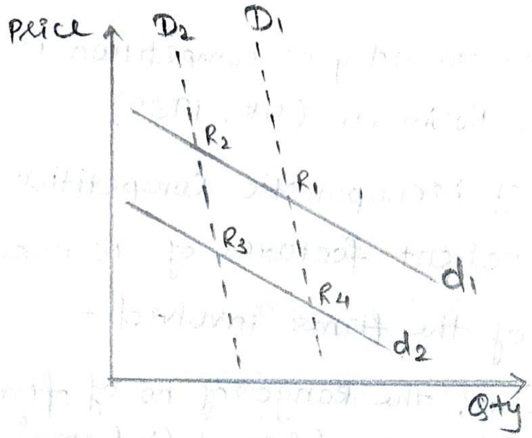
b) According to Chamberlin, when companies deal with similar but differentiated products, they have to inform potential consumers so as to form the mindset of consumers. Hence, they spend on marketing gimmicks like Ads, packaging and service provision. By selling their products, firms hope that eyeballs convert into penny falls. In this ~~marketing~~ selling cost activities are a lifeline as well as a lifebuoy for firms.

c) Firms can indulge in 2 types of competitive strategies:

→ Price cutting strategies
→ Technology enhancement strategies.

3) The Nature of the Demand Curve involved :-

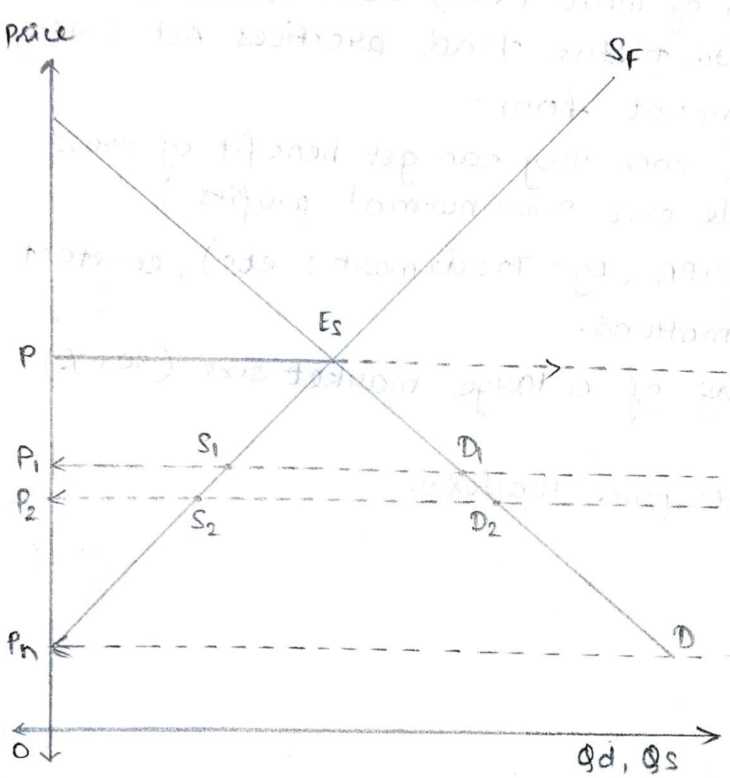
Chamberlin's Analysis discovered that in Mon. Comp. Market, firms encounter 2 types of demand curves :



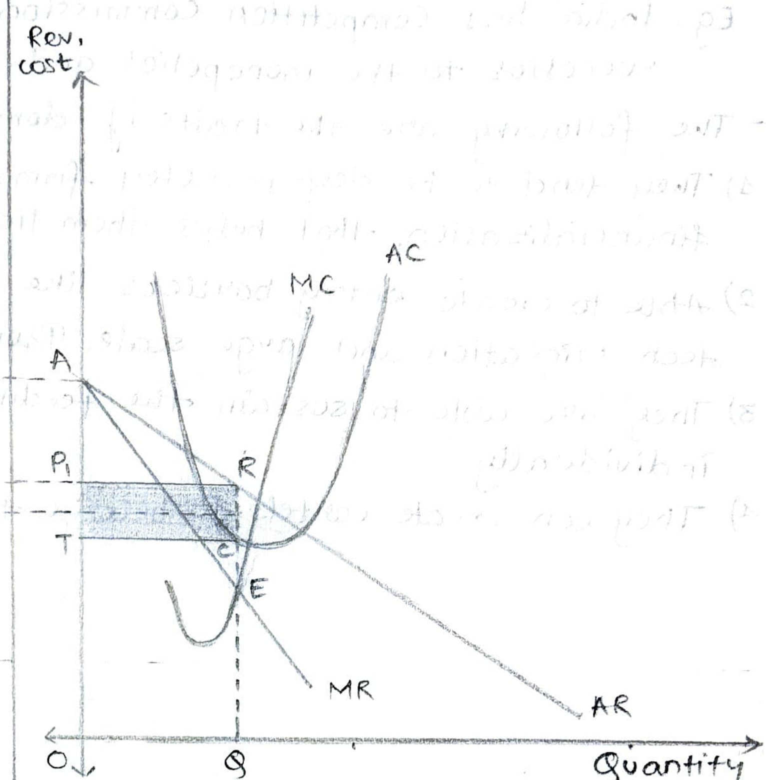
Subjective Demand Curve (individual)	Objective Demand Curve (Group)
• It is the trendline that depicts the commercial aspirations of that firm Eg: When firm lowers its price, it aspires to capture a large market size.	• It is that demand curve that captures the behaviour pattern of all firms involved when firms individually (but collectively) lower the price, the market will increase but one or few firms cannot get the benefit of capturing the entire increment of the Market - The increment would have to be shared among the group members, thus the behaviour pattern of the groups constrains the aspiration of individual firms.
• Chamberlin identified this curve as dd.	• Chamberlin identified this curve as DD.
• It is gently sloped downwards, a fall in price would hopefully lead to a significant increase in demand.	• It is steeply sloped downwards, since the law of Avgs. would force the increment in market to be much lesser than the aspired one.
• When the dd curve shifts downwards, it indicates that the concerned firm is lowering price and vice versa.	• When DD curve shifts to the right it means that the firms are existing the market. When DD curve shifts to the left it means new firms are entering the market.

OLIGOPOLY

Shared Market



Dominant Firm



- In figure 1, the Demand curve (D) shows the demand that the consumers exhibit and all the firms (smaller and dominant) would like to cater to.
- The $P_n - S_F$ curve represents the supply function of the smaller firms. Hence, E_S becomes the equilibrium from which the smaller firms operate.
- From E_S, it will be noticed that the average price charged by the smaller firms is OP and the supply size of smaller firms is $P - E_S$ units.
- Once the dominant firm realizes that smaller firms on an average would be charging a price OP would be equal to OA in figure 2. The dominant firm would begin operating from that point.
- Hence, the financial functions (AR, MR, AC, MC) will be based on the position of point A. Since this firm is the dom. firm, it can opt for profit maximization and so operate from point E and produce OQ units and charge a price OP_1. (NOTE: $OP_1 < OP$)
- The dom. firm passes on its low price into the market. Since the DF has lowered the price, the demand increases from $P - E_S$ to P_1D_1.
- The market share of smaller firm is P_1S_1 and ∴ DF's market share is S_1D_1
- In order to increase its market share further, the DF would continue to indulge in the price cutting strategy by even sacrificing max. profits.

- In theory the DF could continuously cut prices till OP_n and monopolize the market space. However all economies across the globe have put legislations in place to prevent such monopolization from taking place.
Eg: India has Competition Commission of India (CCI) 2003 which is successor to the monopolies and restrictive trade practices Act (1969)

- The following are the traits of dominant firm:

1) They tend to be deep pocketed firms since they can get benefit of cross financialization that helps them tide over sub-normal profits.
2) Able to create entry barriers like IPR (Eg: Trademarks etc), constant tech Migration and large scale Promotions.
3) They are able to sustain the feeding of a large market size (30+%) individually.
4) They can create cartels to become the price leaders.